

Course 059250 - Soft Robotics

Politecnico di Milano | June 2026

Bio-Inspired Soft Gripper

Underactuated Morphology for Delicate
Agricultural Manipulation

GROUP 7:

Yigiter Tuncer

Lim Thierry

Yuxuan Lu

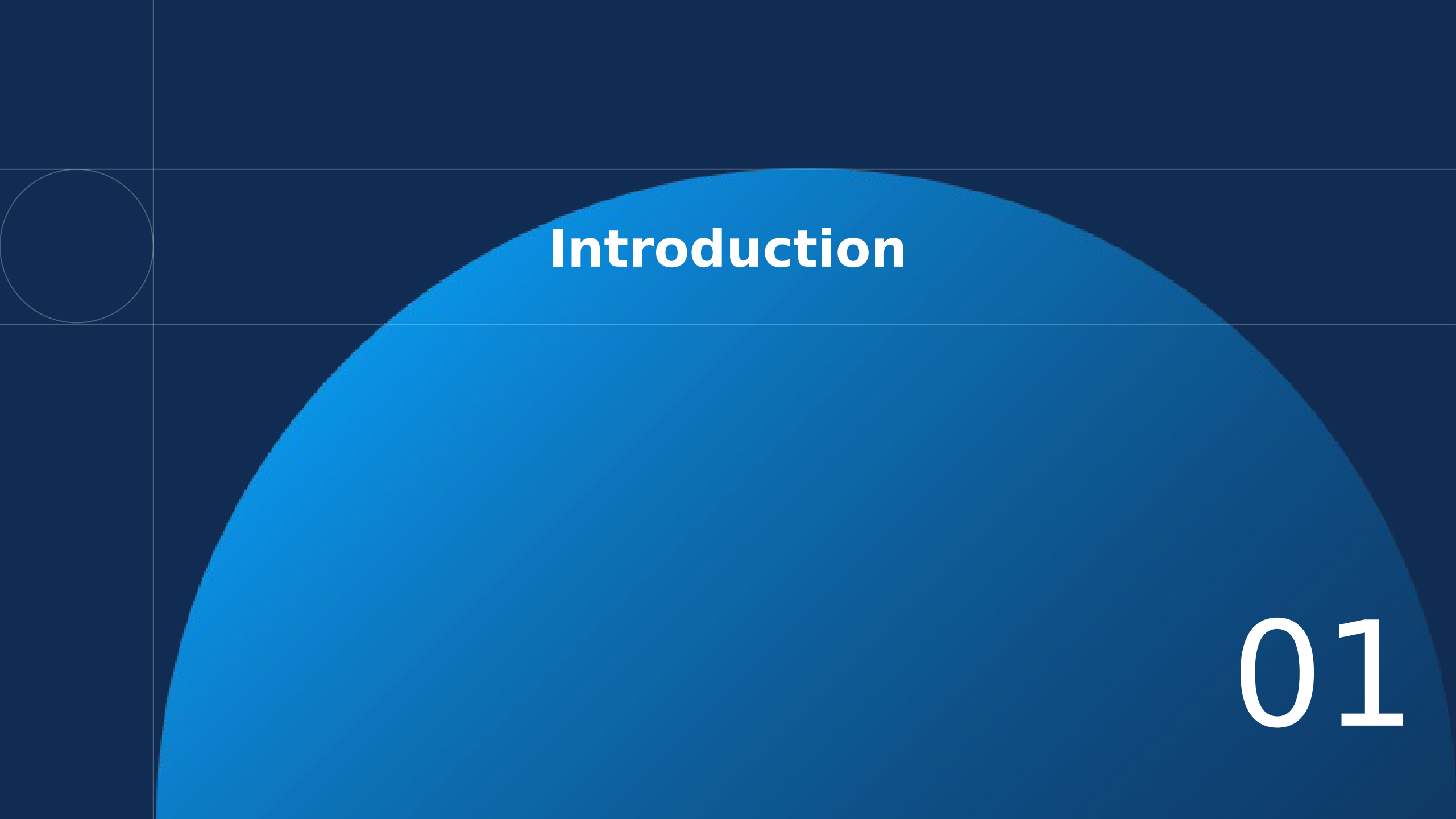
Sajedah Mazinani

Xiheng Li



Index

- 1. Introduction**
- 2. literature review**
- 3. CAD Design**
- 4. Circuit**
- 5. Material**
- 6. Results**
- 7. Possible Improvements**



Introduction

01

Context & Problem

The Mushroom Challenge

Our target application requires the manipulation of hyper-fragile geometries. Traditional rigid end-effectors fail due to two primary biological constraints:

- 01 Low Yield-Stress Threshold**
Surface pressure exceeding specific limits causes immediate cellular rupture and enzymatic browning.
- 02 Geometric Heterogeneity**
Asymmetrical topologies create unpredictable point-contact stresses.

*"The failure of rigid automation in agriculture is not a speed problem, it is a **compliance** problem."*



Motivation & Inspiration

CONTEXT & TARGET

Automation in agriculture demands high delicacy for mushroom harvesting.

THE PROBLEM

Irregular, asymmetrical shapes. Extremely sensitive to point forces (bruising/damage).

INDUSTRY STANDARD

Prone to clogging from soil/spores; localized negative pressure causes cap bruising.



OUR INSPIRATION

Moving from top-down suction to a 3-fingered wrapping motion for uniform pressure distribution.

Kinematic Goals & Criteria

m
e
m
or
y

Passive Adaptation

Eliminating the need for expensive sensor feedback loops by utilizing **Morphological Computation**. The material *is* the controller.

co
m
pr
es

Pressure Homogenization

Achieving uniform stress distribution across the cap surface to maintain **Structural Integrity** of the biological tissue.

set
tin
gs
_in
pu
t_c
o

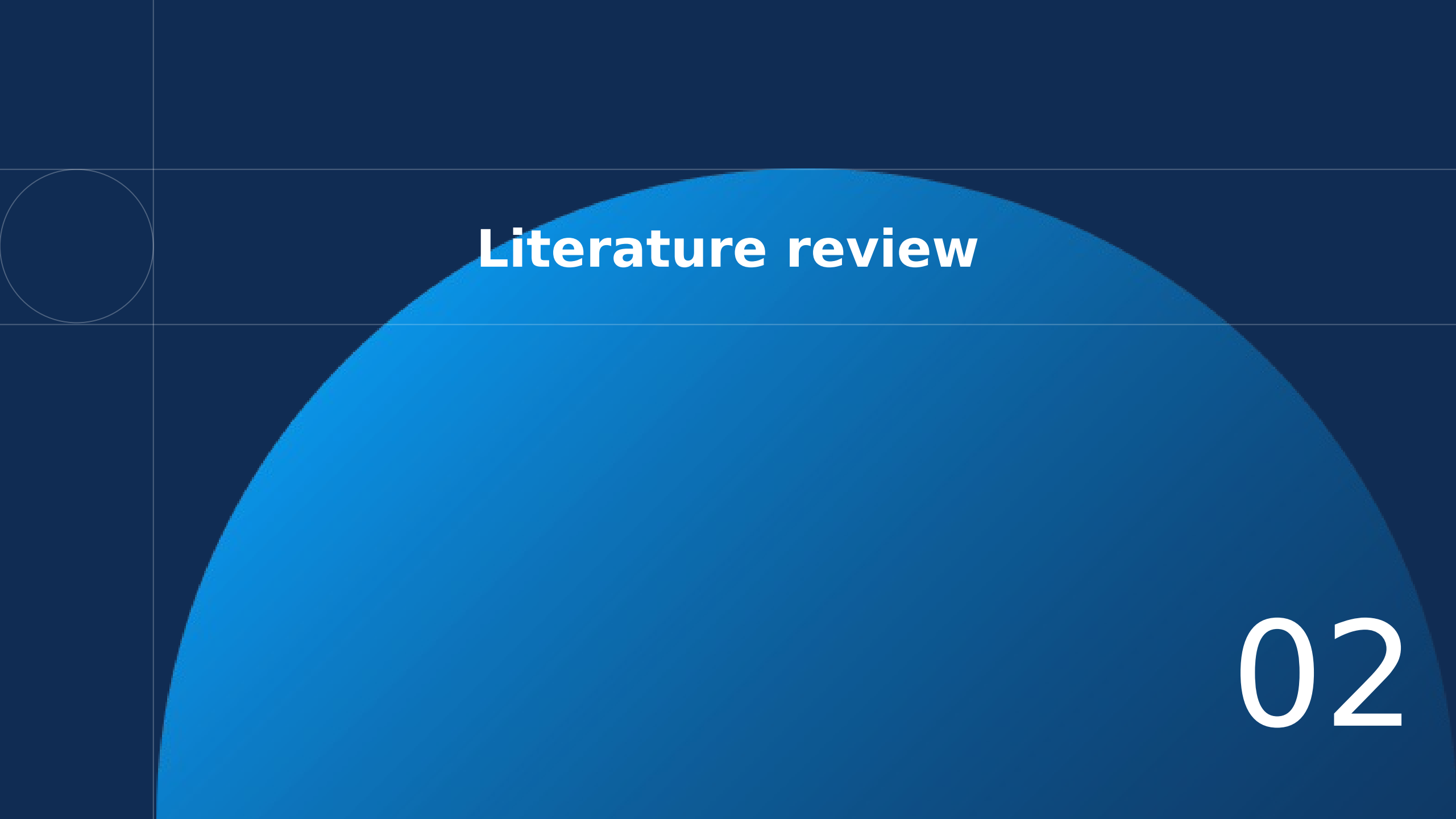
Underactuated Simplicity

Reducing hardware complexity to a single motor, ensuring **Synchronized Flexion** with minimal electronic latency.

se
arc
h

Research Question

How can an underactuated soft morphology synergize force-isotropic tendon routing with optimized surface tribology to achieve a secure, zero-bruise enveloping grasp on highly sensitive asymmetrical targets?



Literature review

02

SOTA: Mushroom Other Approaches

Hat approach



[3] Suction



[4] Soft finger

Stem approach



[5] Cutting finger



Context Note

*These methodologies provide several different **advantages** and **drawbacks**, mainly chosen according to the type of mushroom*

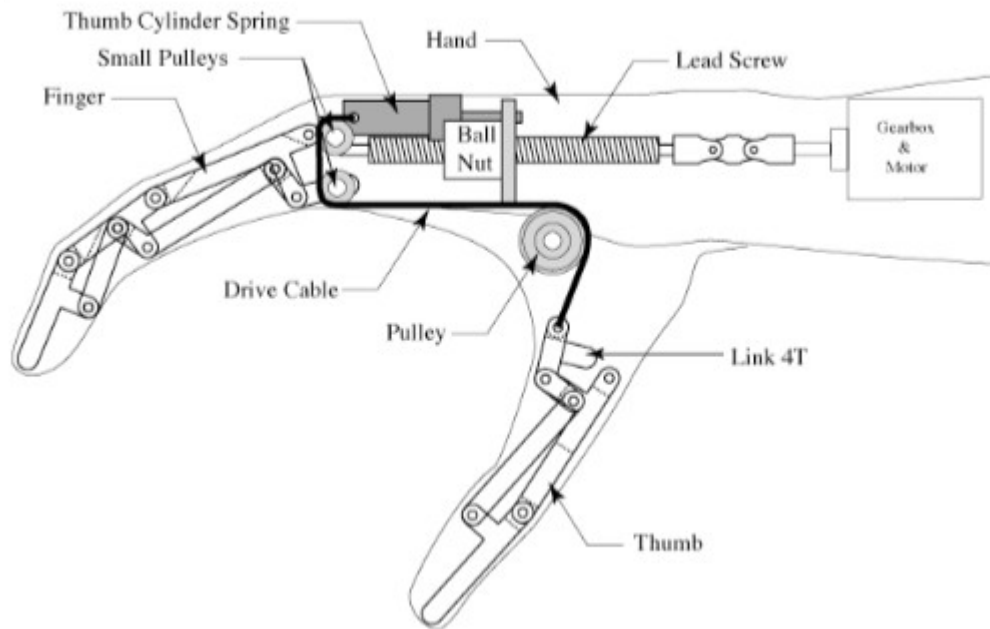
[3] Koirala, B. et al., "A Hybrid Three-Finger Gripper for Automated Harvesting of Button Mushrooms", Actuators, Vol. 13, No. 8, 2024.

[4] Ceres Agri-Tech, "Ceres Agri-Tech to fund mushroom-picking robots", The Engineer, 2024.

[5] Koirala, B. et al., "Robotic Button Mushroom Harvesting Systems: A Review of Design, Mechanism, and Future Directions", Applied

Actuation between fingers with Tendon (Serial)

One cable that passes **sequentially** throughout all the fingers.



[6] Route of drive cable

Pros:

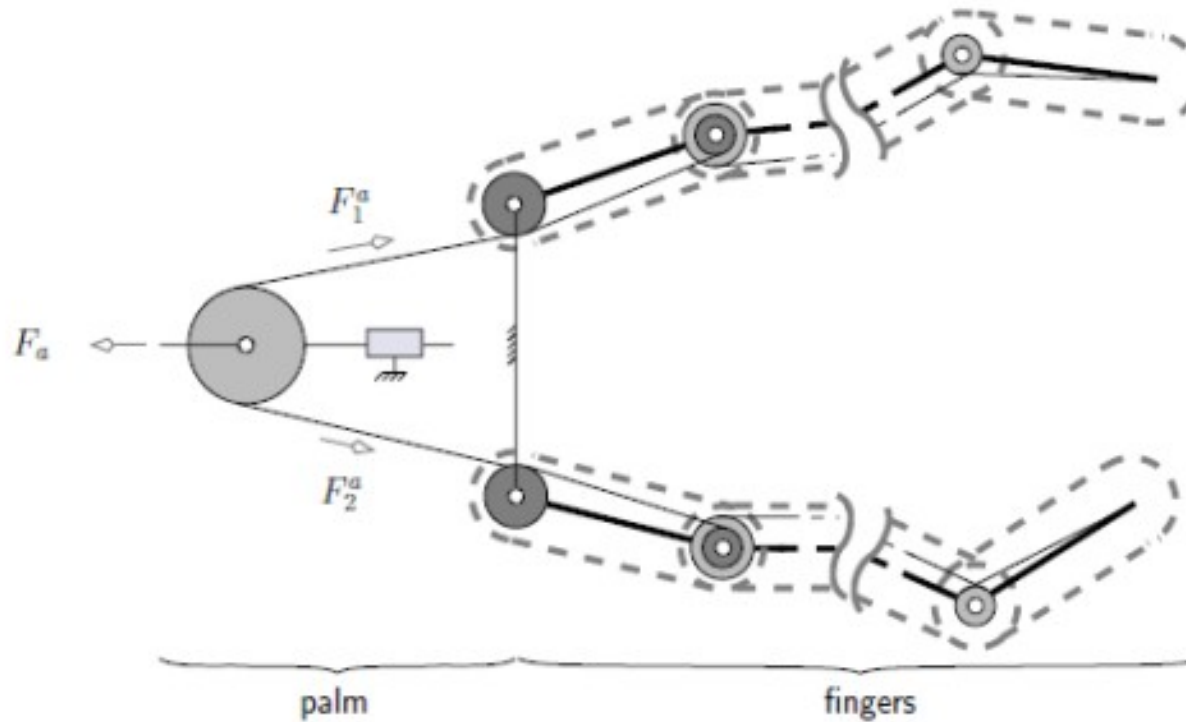
- Highly **adaptive** to asymmetrical objects.

Cons :

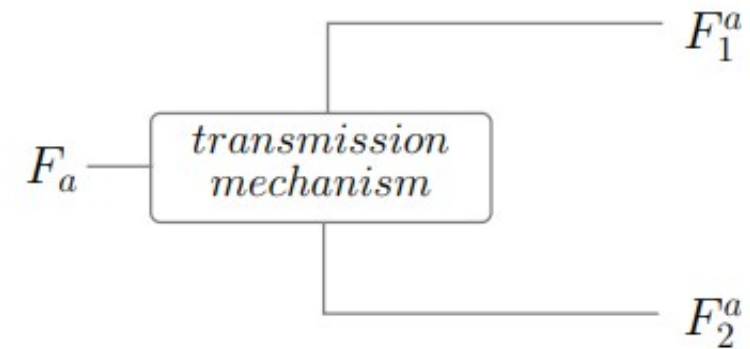
- Distal force drop
- **Angle** of finger might be **different** and more harder to control

Actuation between fingers with Tendon (Parallel)

Often use **Multiple Pullers** or **Whiffletrees** in order to distribute the force angle.



[7] Movable pulley driving two underactuated fingers

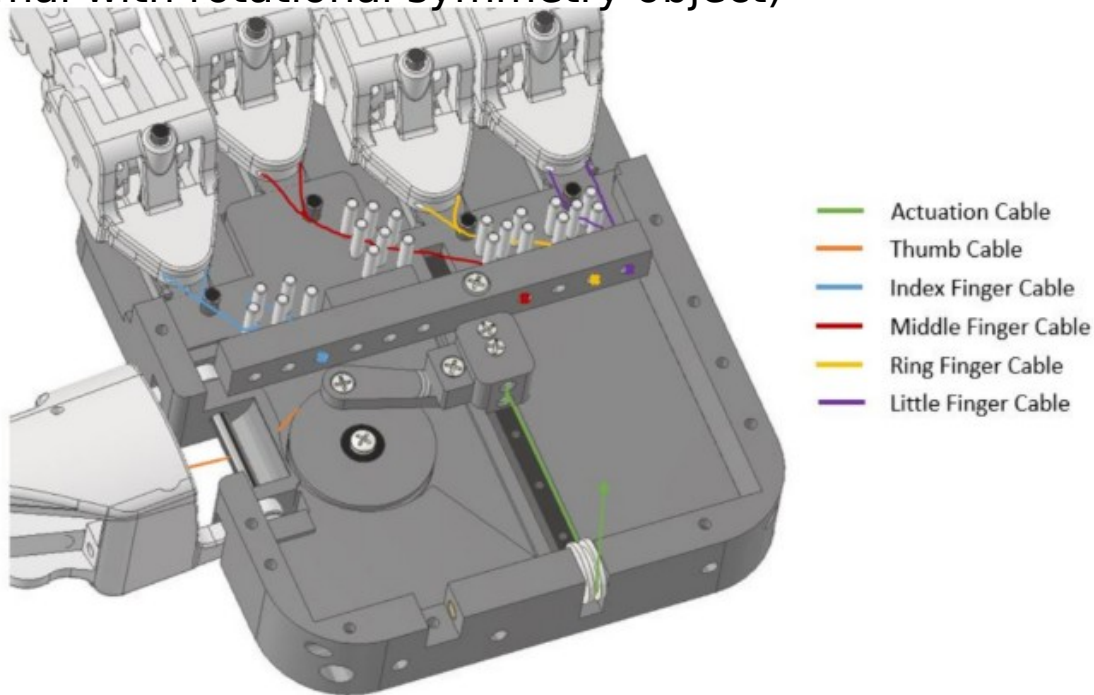


[7] Force transmission system

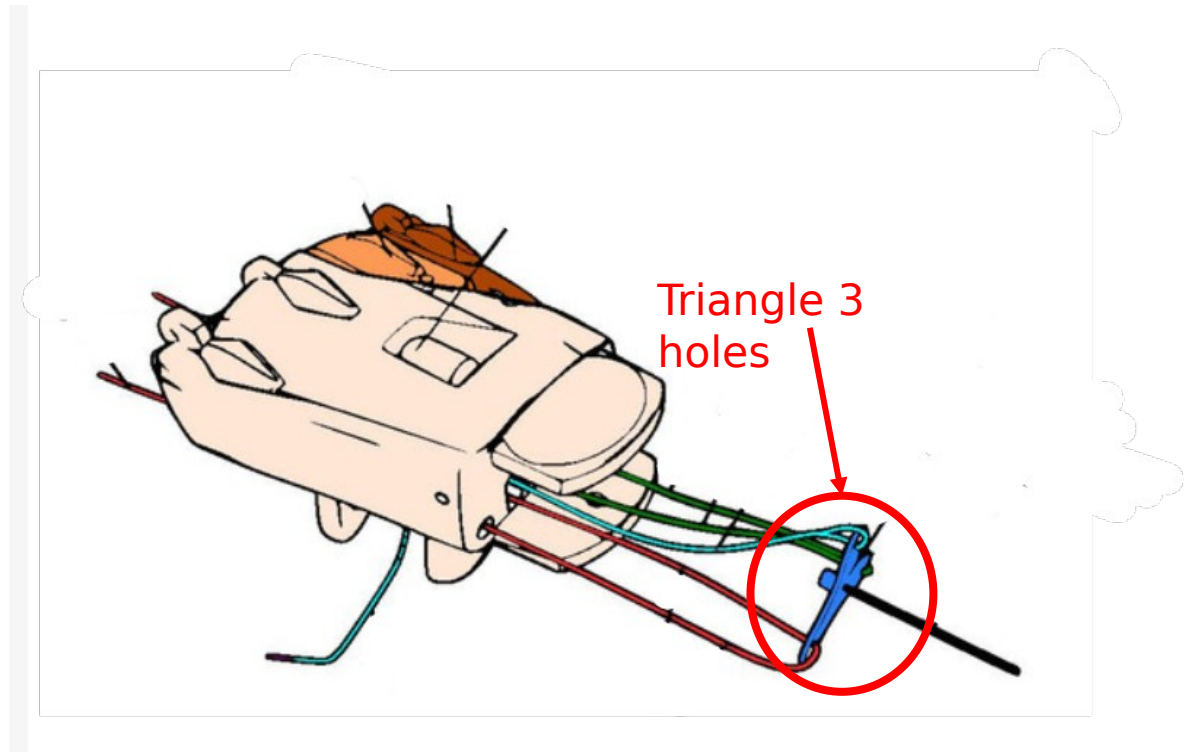
Actuation between fingers with Tendon (Parallel)

Some whiffletrees bypass multiple pulleys, routing all cables into a **single actuation system** to maintain force isotropy.

(Optimal with rotational symmetry object)



[8] Power transmission mechanism



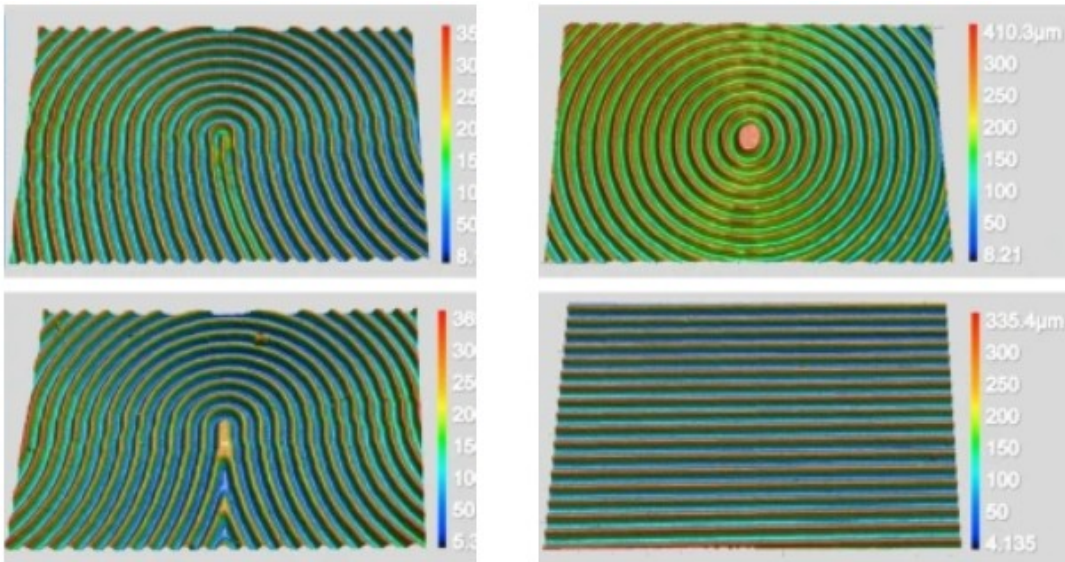
[9] Triangle Whiffletree

[8] Nikafrooz, N. & Leonessa, A., "A Single-Actuated, Cable-Driven, and Self-Contained Robotic Hand Designed for Adaptive Grasps", Robotics, Vol. 10, No. 4, 2021.

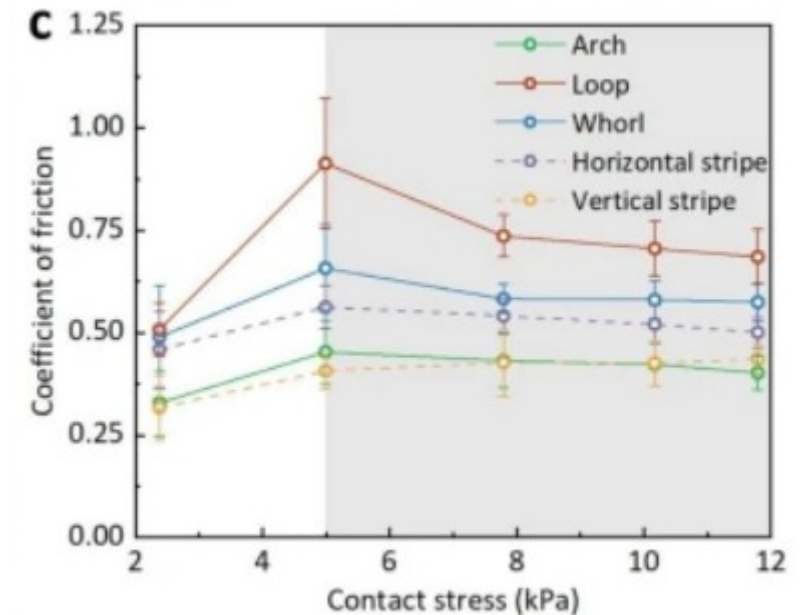
[9] Satriawan, A. et al., "Karla: A Simple and Affordable 3-D Printed Body-Powered Prosthetic Hand with Versatile Gripping Technology", Designs, Vol. 7, No. 2, 2023.

Tribology (idea)

We may be interested in **increasing the friction by maximising the surface area**; this can be done by using **patterns**



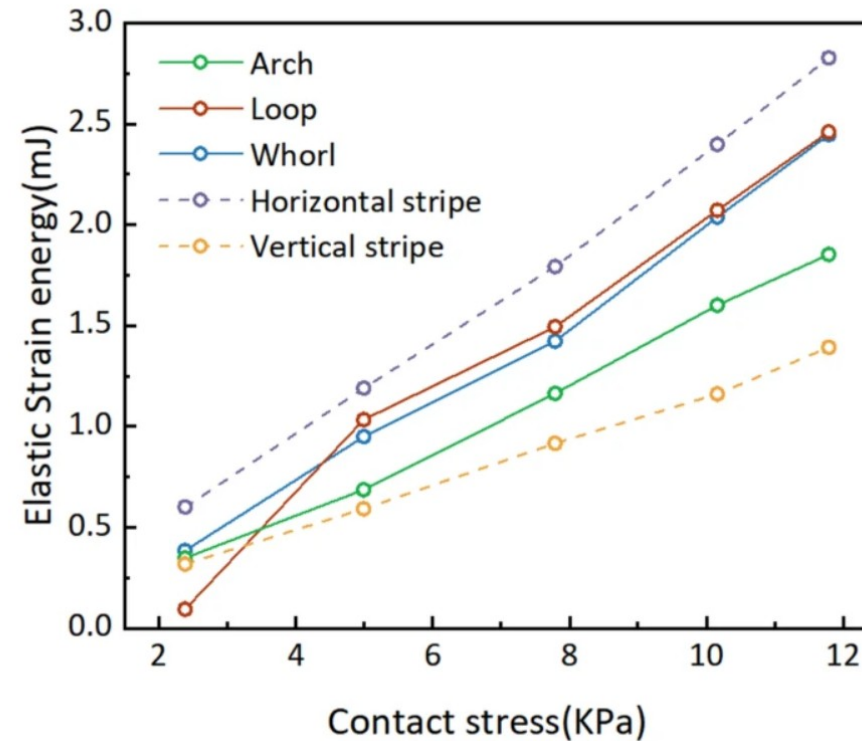
[10] Example of different tribo-skin pattern



[10] Friction coefficient for different pattern

Tribology (idea)

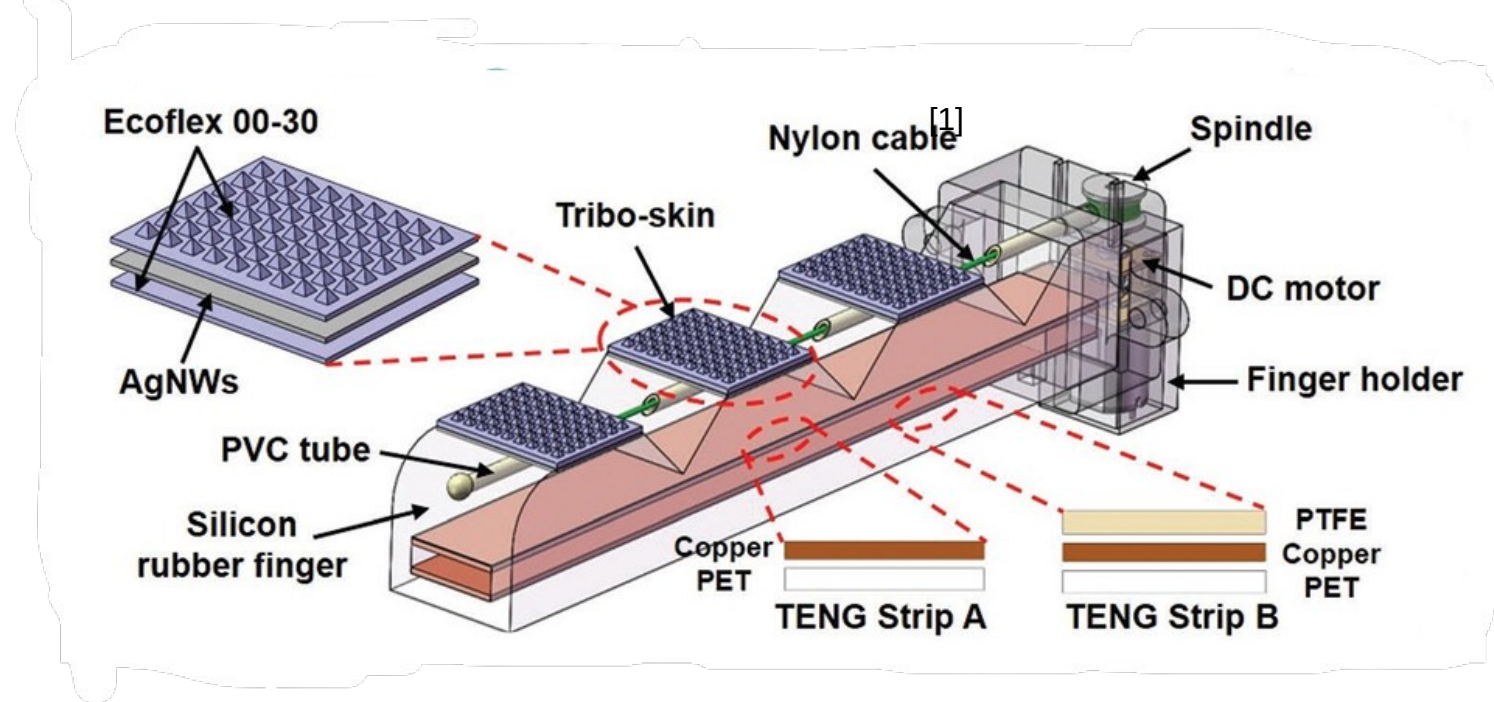
Friction isn't the only criterion for the pattern. A good design can allow for **anchoring**.



[10] Elastic Strain Energy for different pattern

Tribology (design inspiration)

The final design is mainly inspired by another paper that maximise sensor detection by maximising surface contact with **pyramidal shape** (tribo skin)



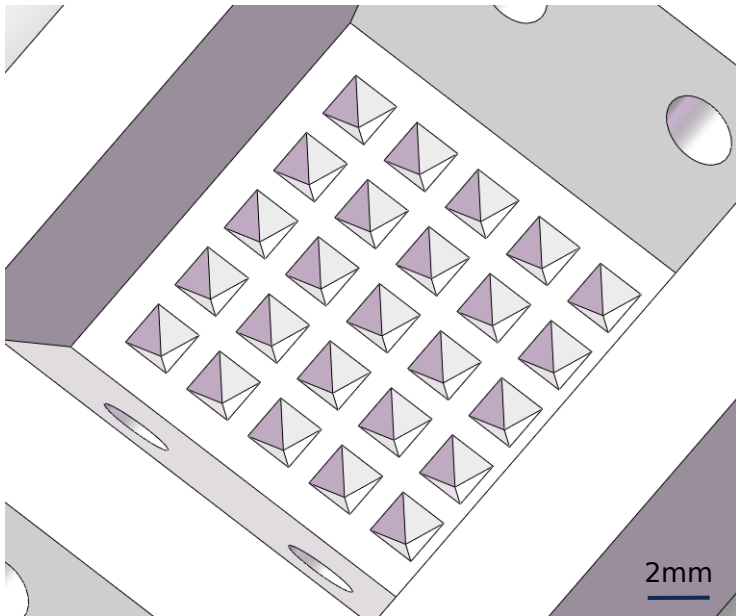
[11] Movable pulley driving two underactuated fingers



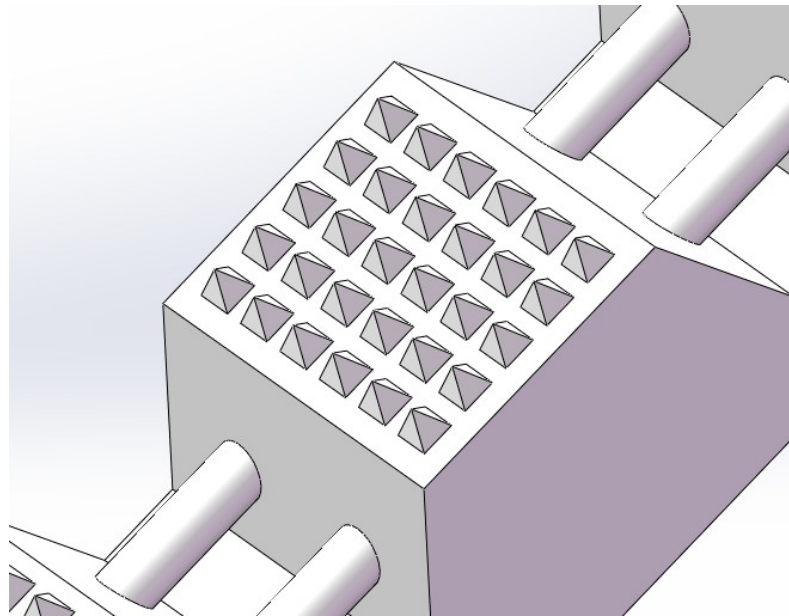
CAD Design

03

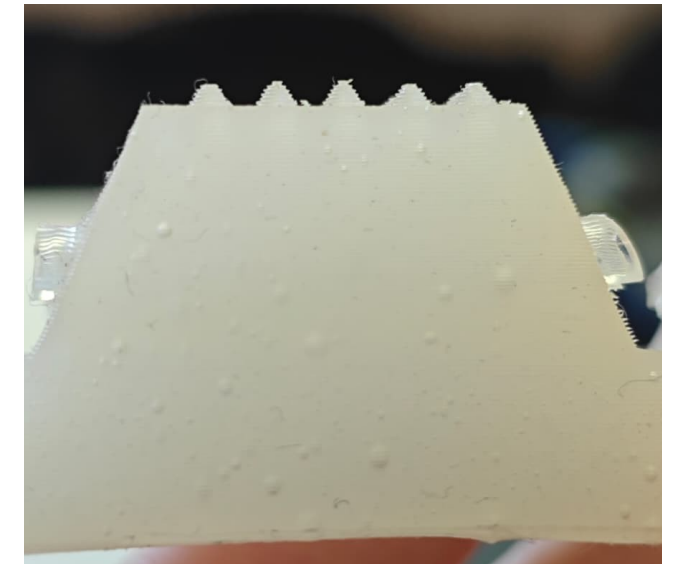
CAD Design - Tribo skin (pattern)



Mold

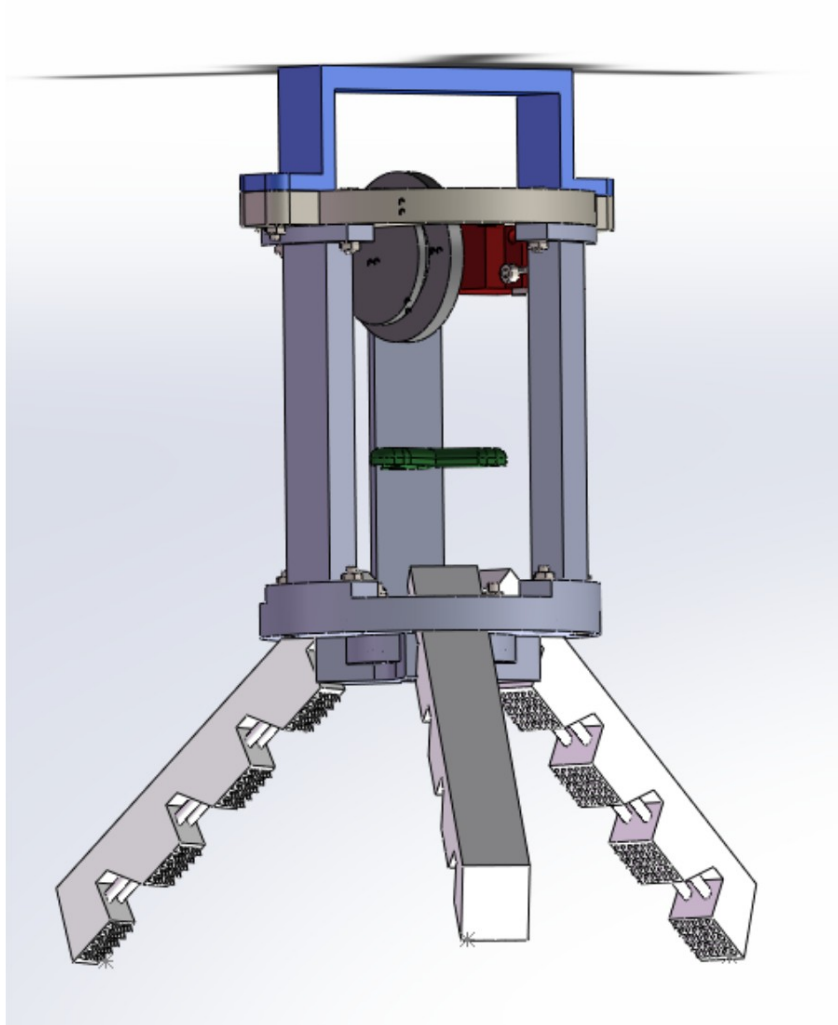


Casting(ideal)



Casting(real-life)

CAD Design - Whole Mechanism

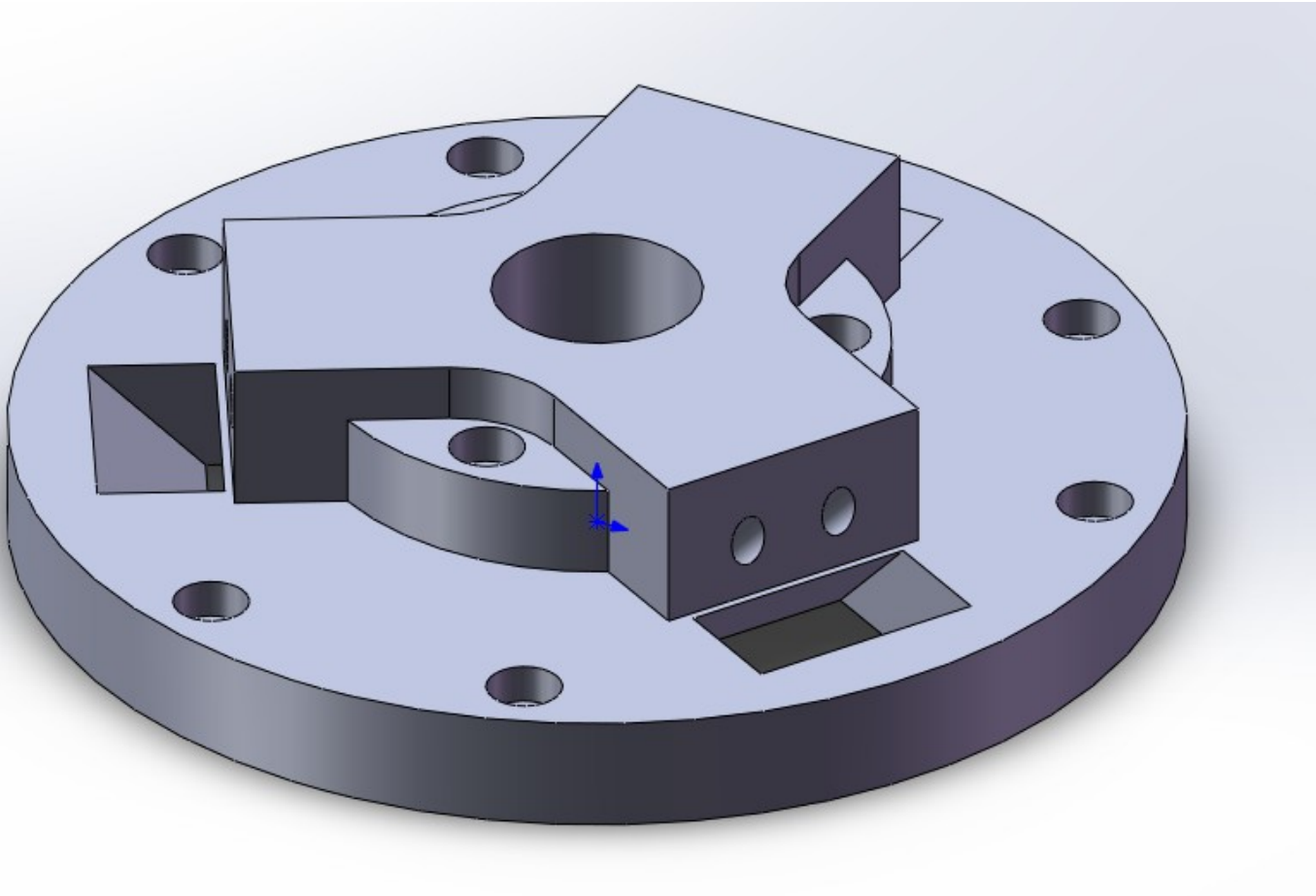


Whiffletrees Mechanism

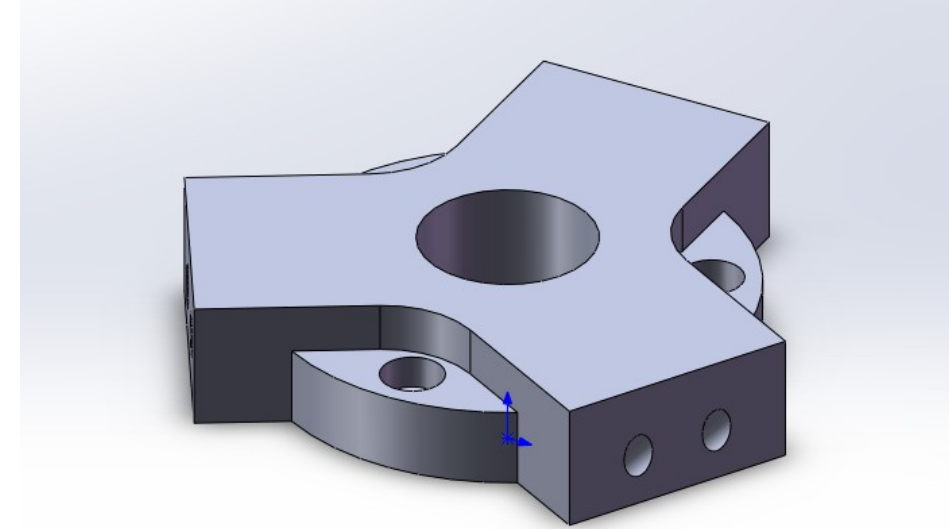
Including:

- Base
- Small Plane
- Pulley

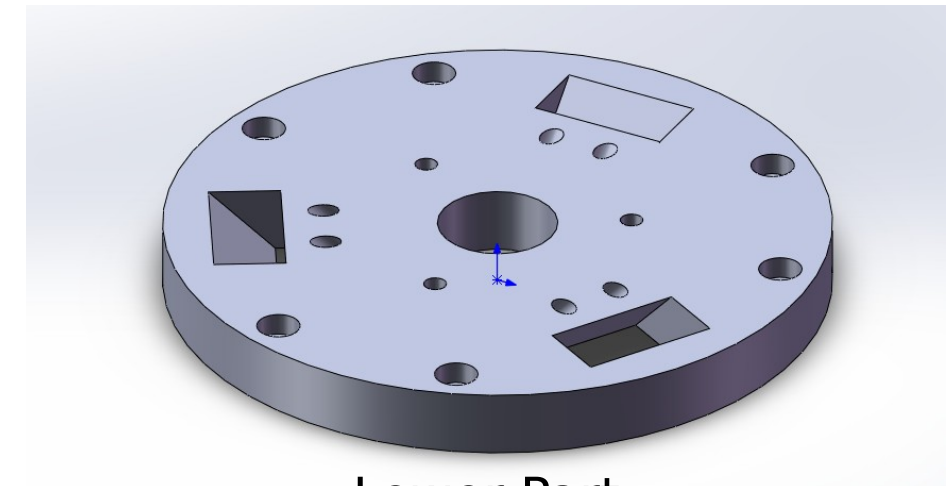
CAD Design - Base



Whole Base

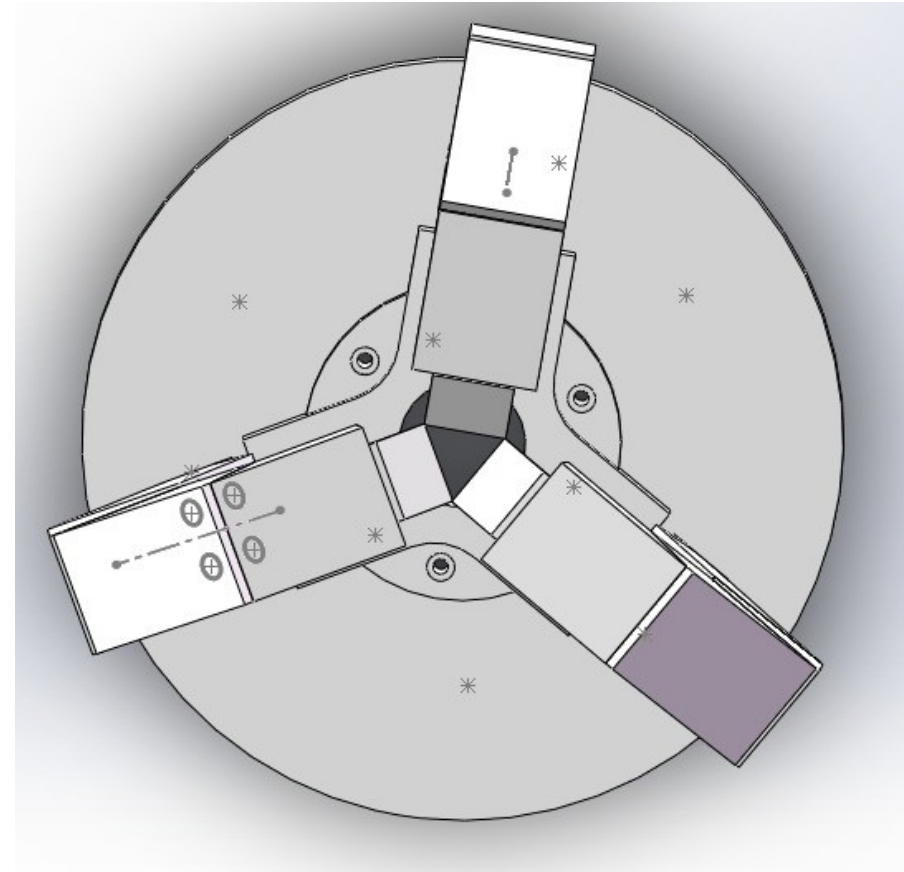
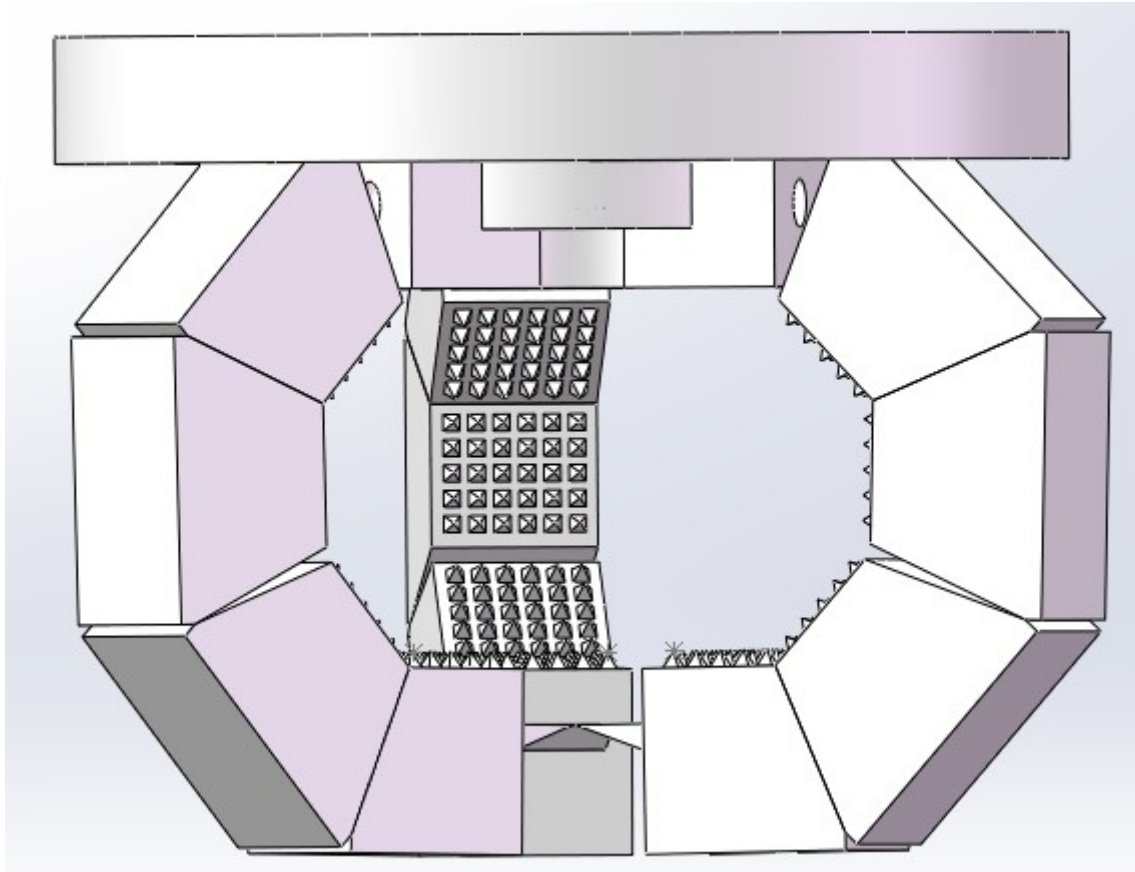


Upper Part



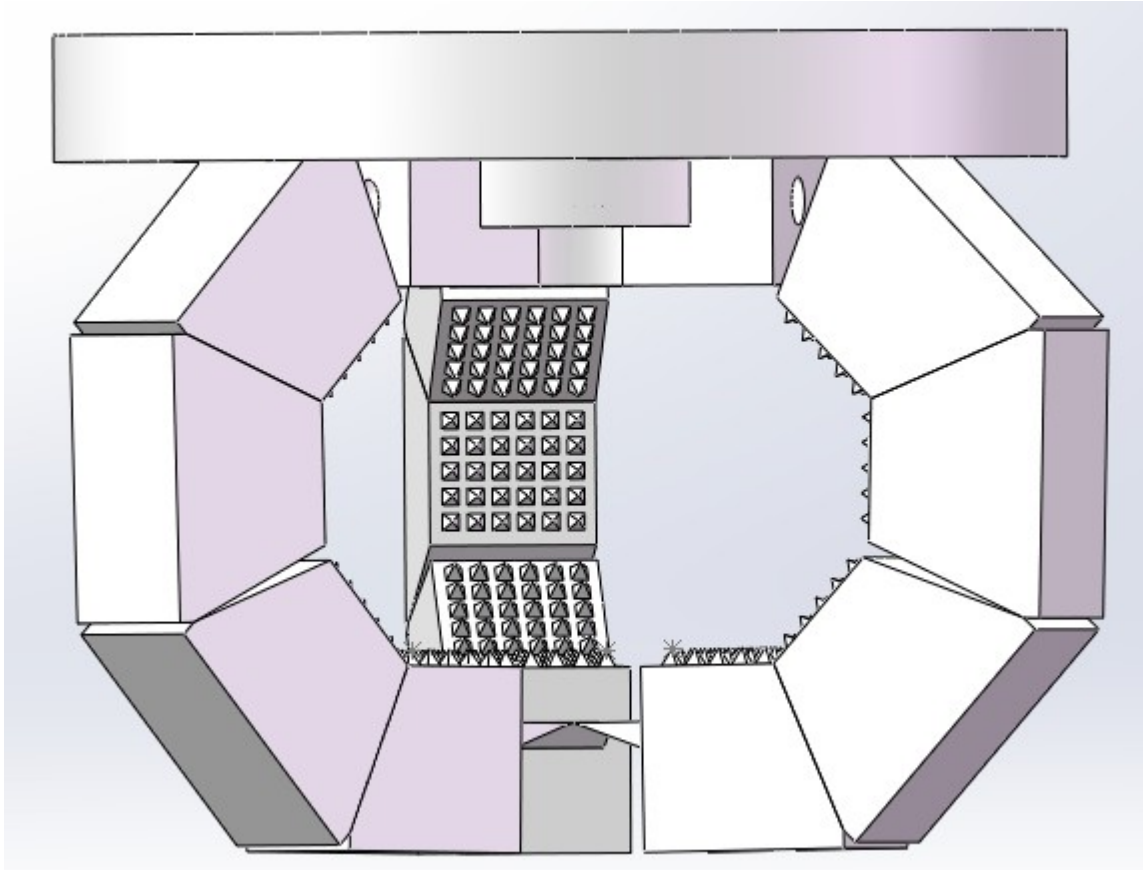
Lower Part

CAD Design - Finger (1st Version)

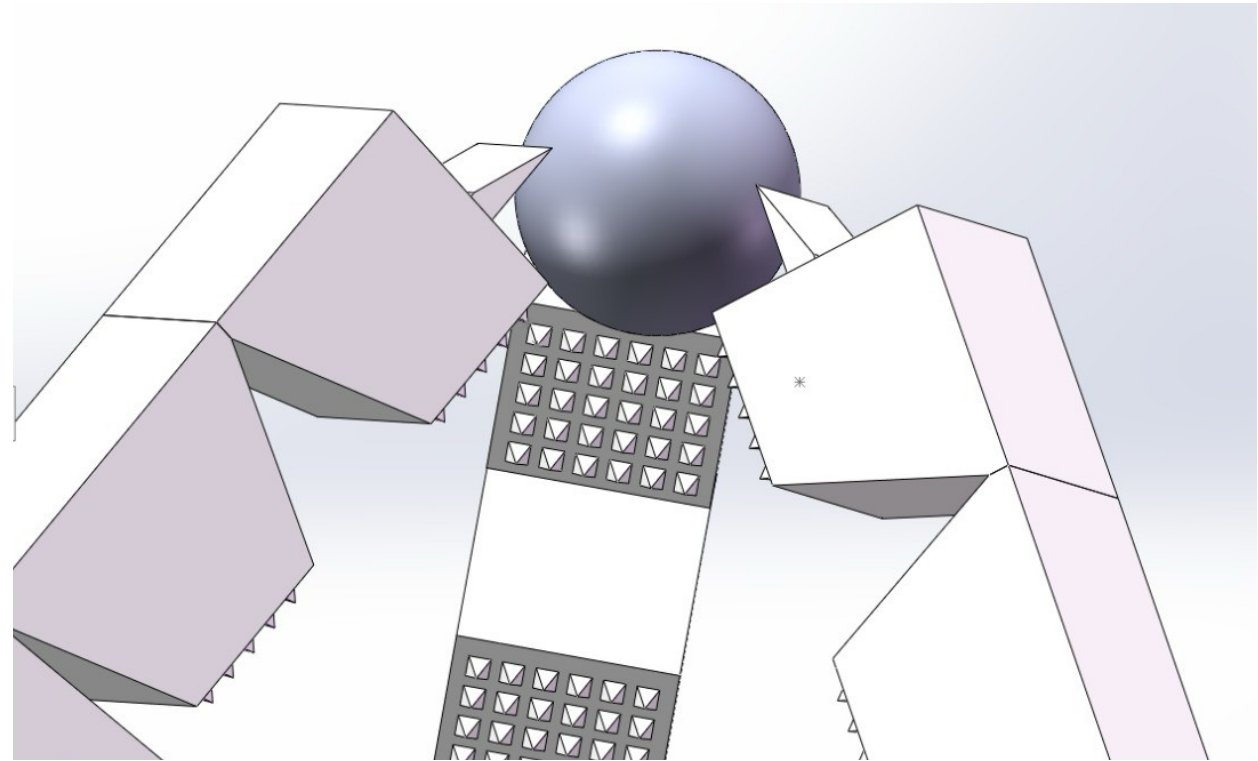


Fully pulled Simulation

CAD Design - Finger (1st Version)



Fully pulled Simulation

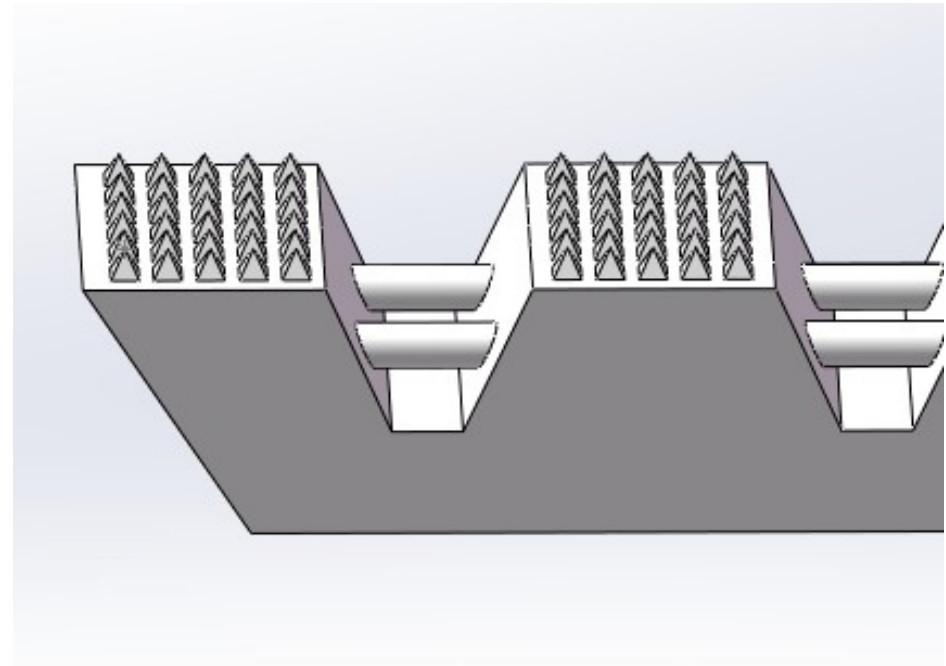


Grabbing Small Object

CAD Design - Finger (Final Version)

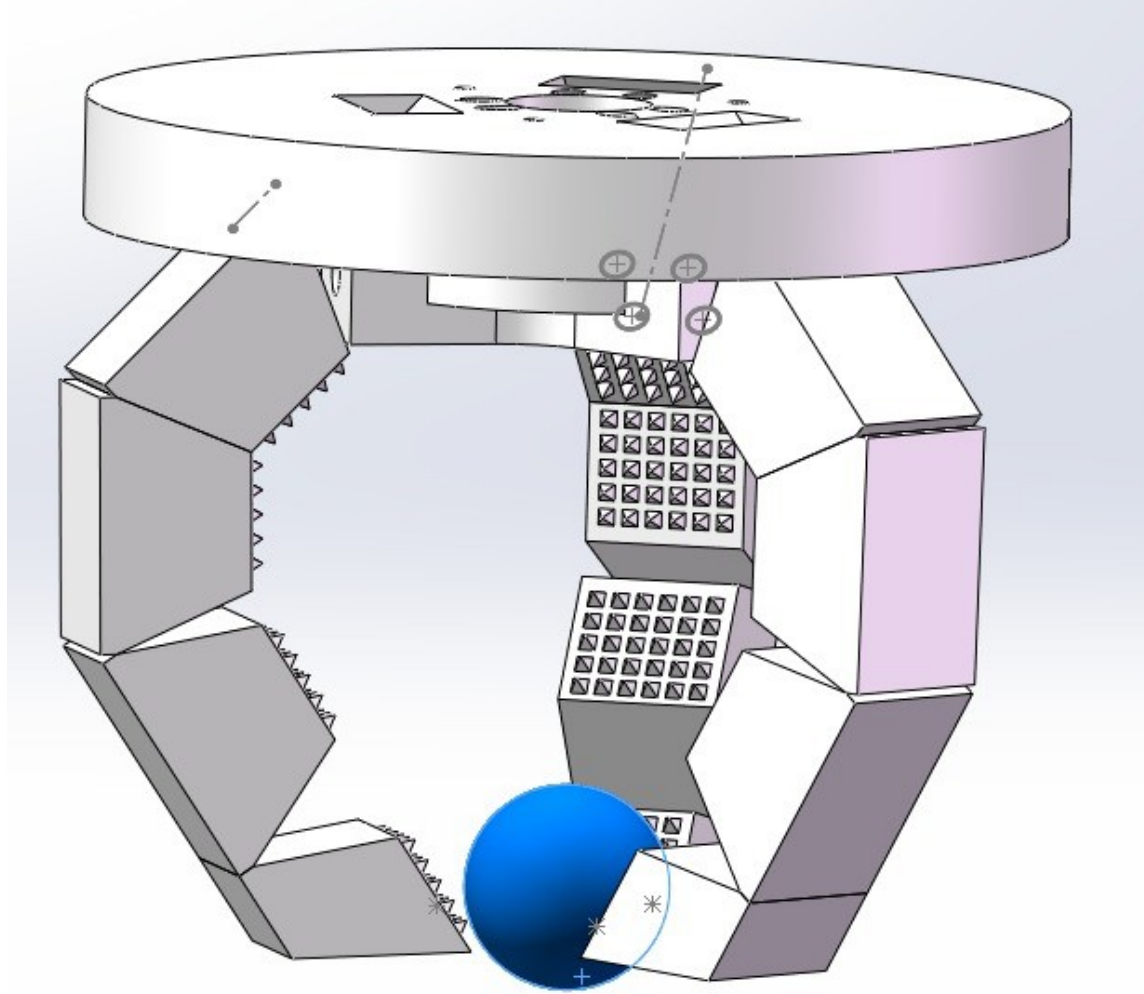


Real Finger



Final Finger

CAD Design - Finger (Final Version)

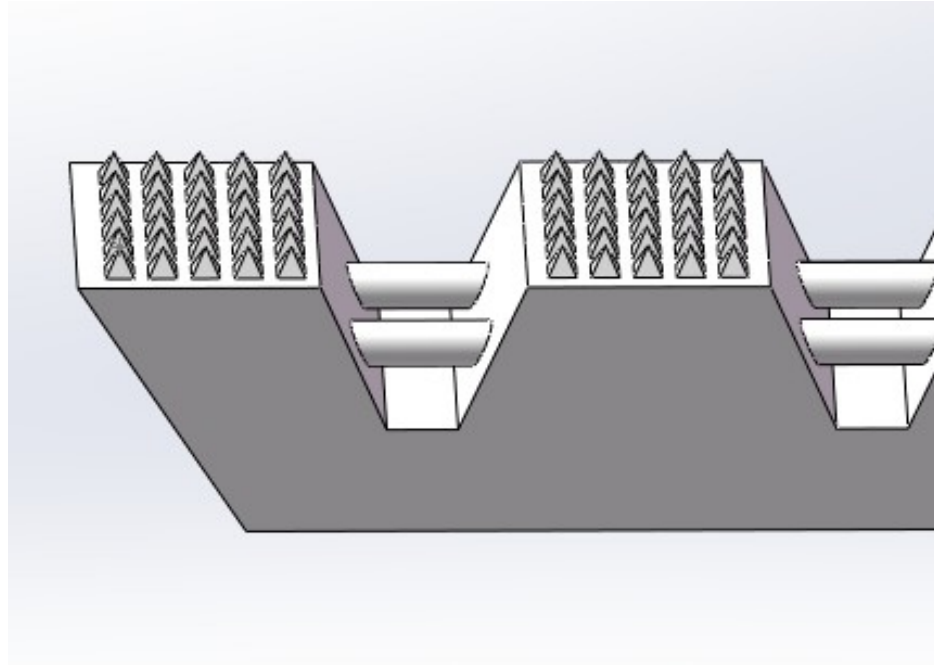


Fully pulled Simulation

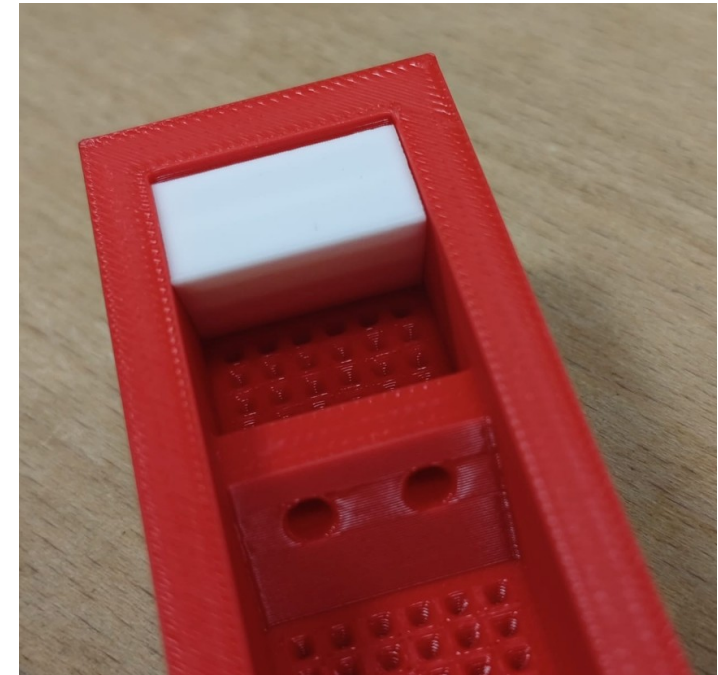


Reality

CAD Design - Mold

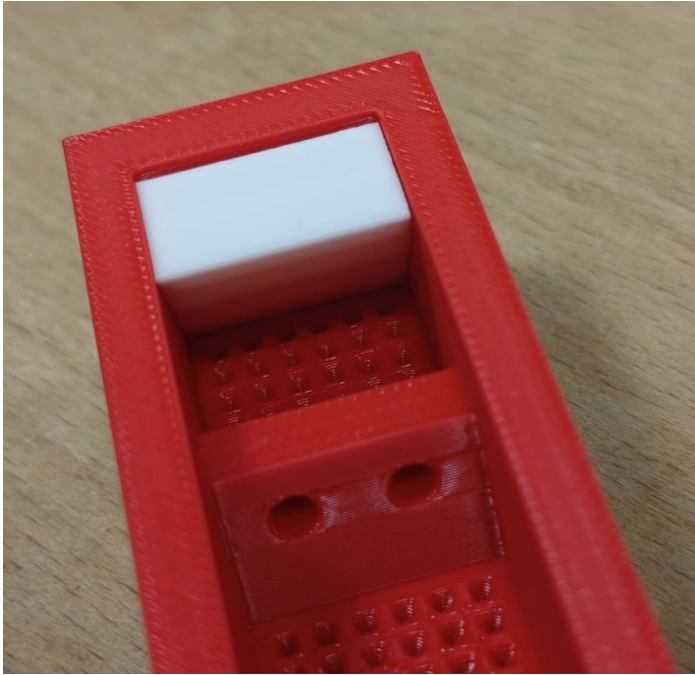


Final Finger

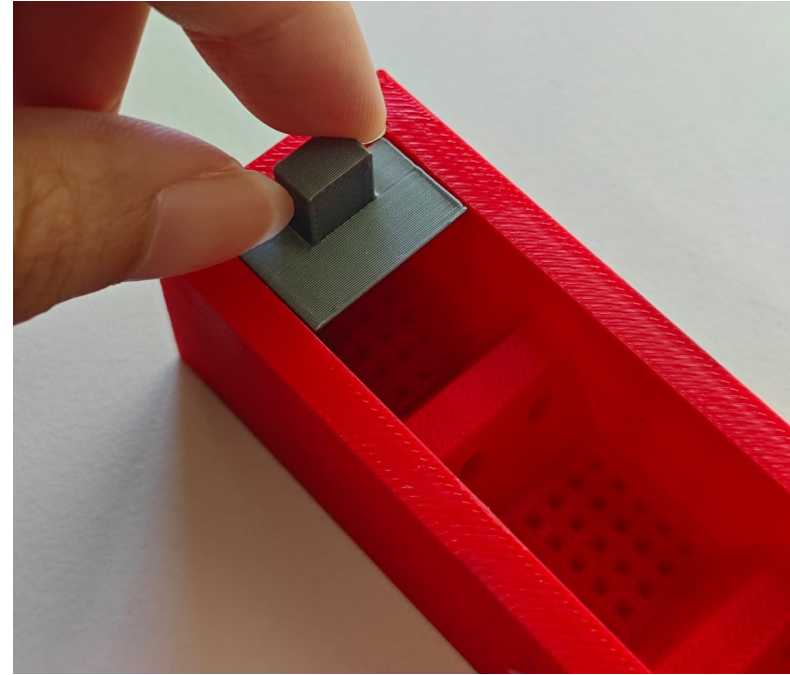


1st Design

CAD Design - Mold

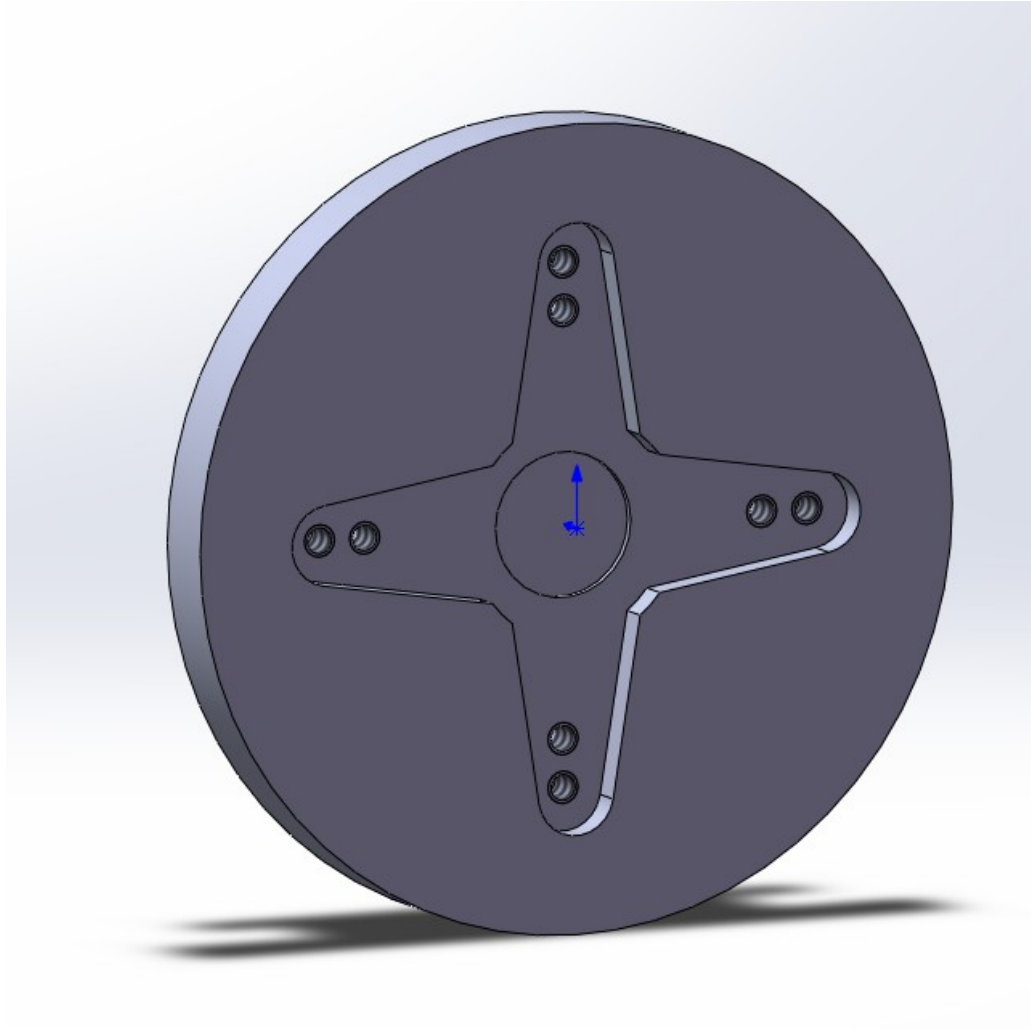


1st Design

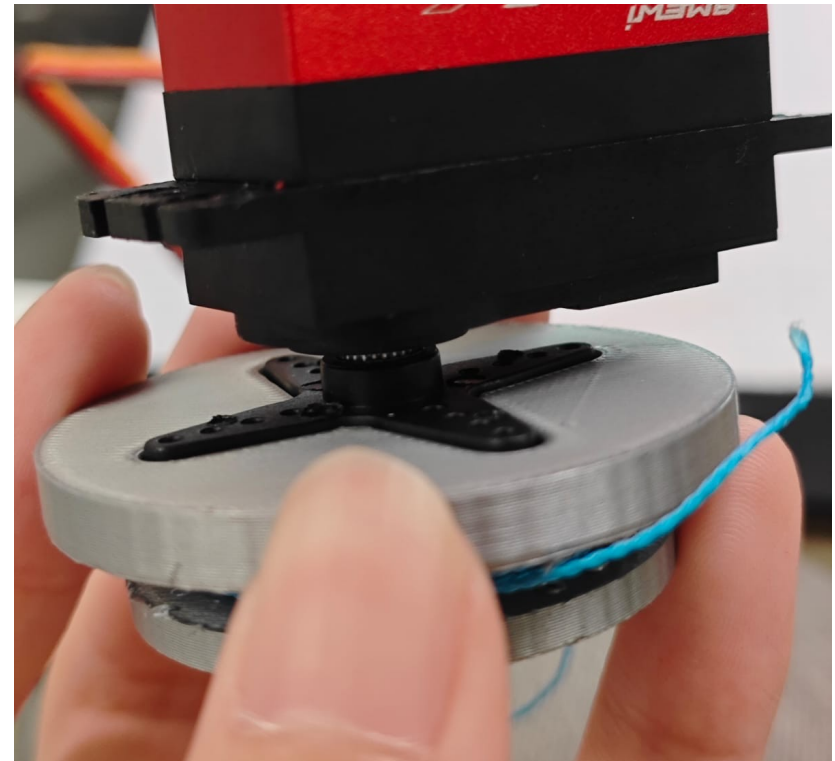


Final Design

CAD Design - Pulley

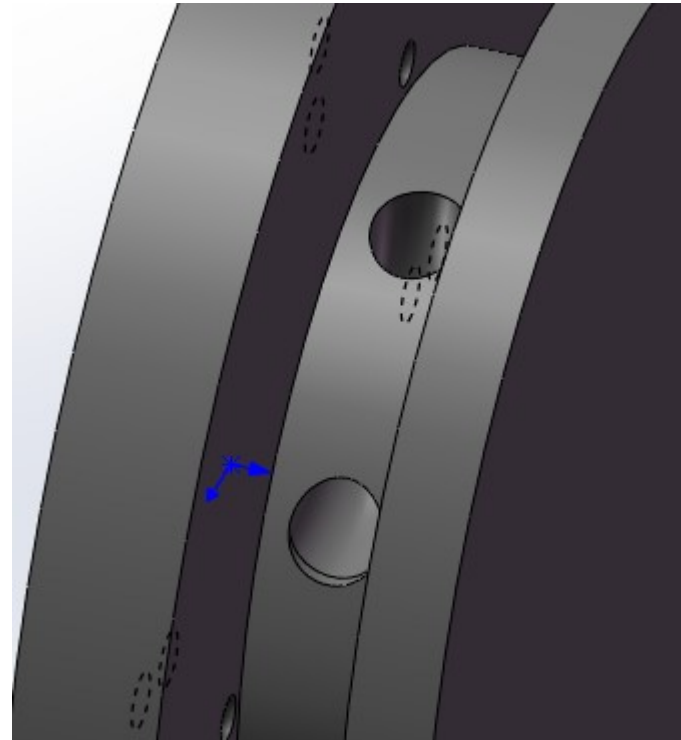


Pulley



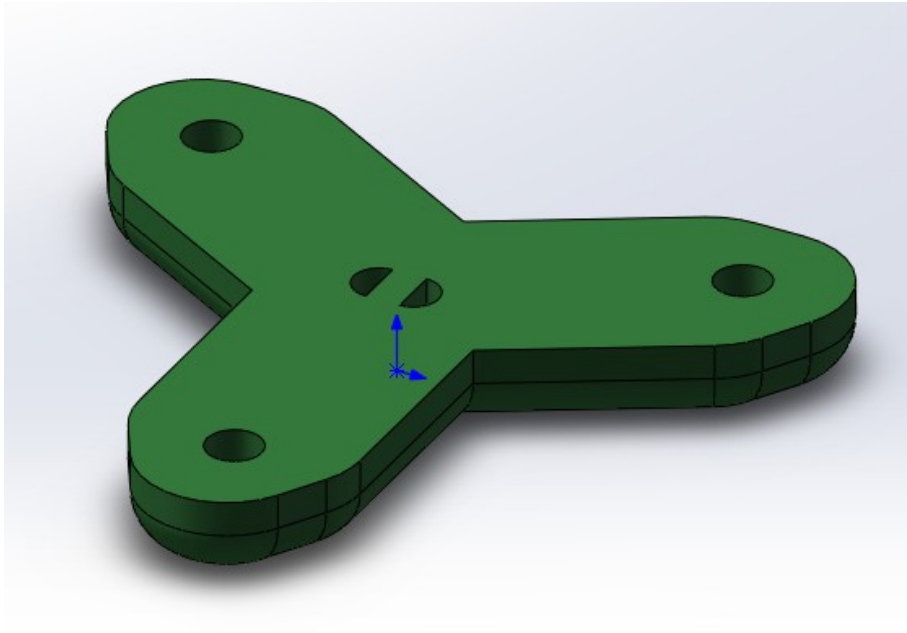
Reality

CAD Design - Pulley

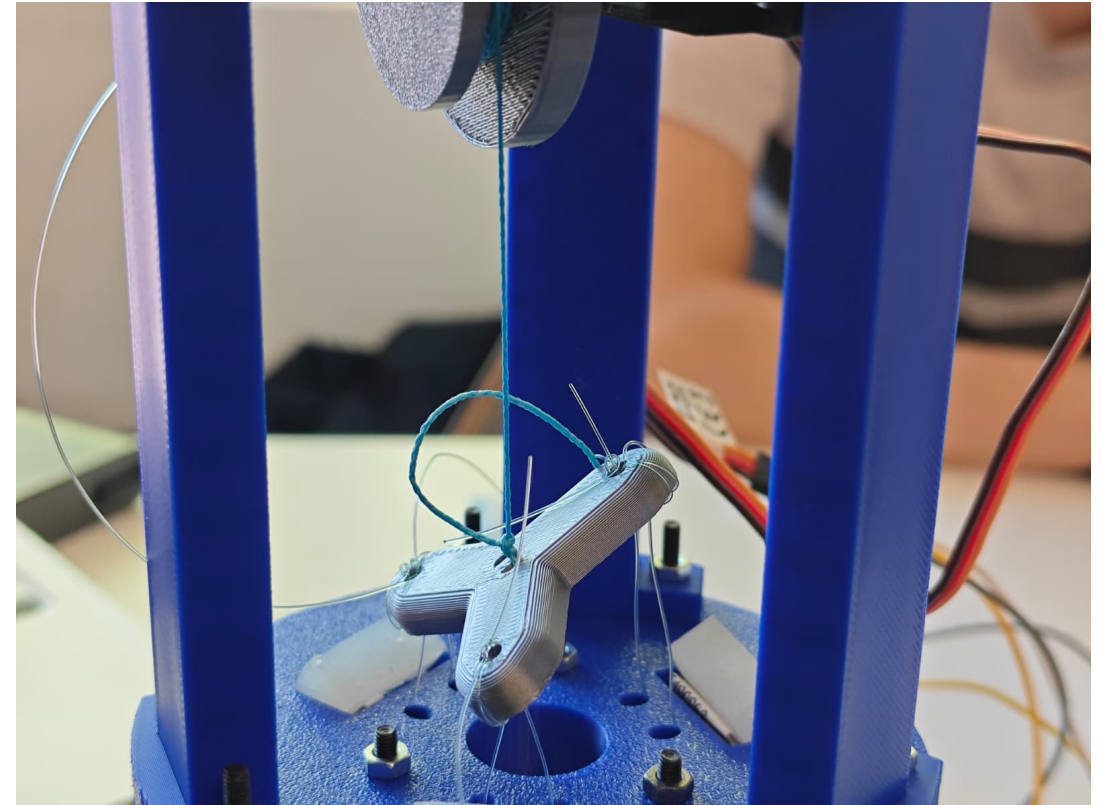


Pulley

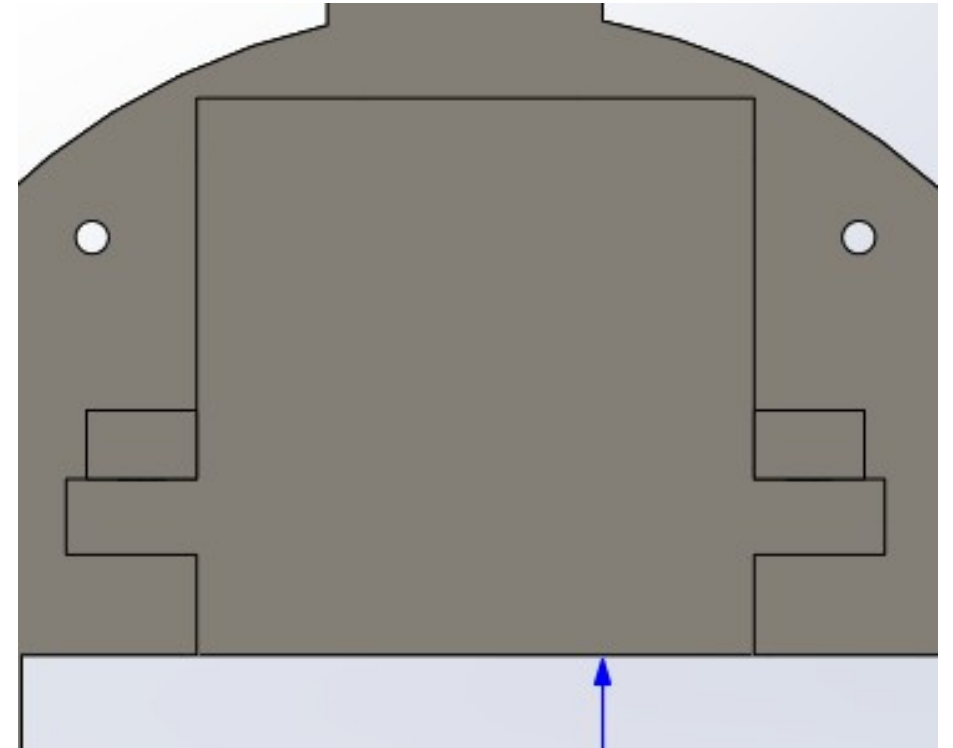
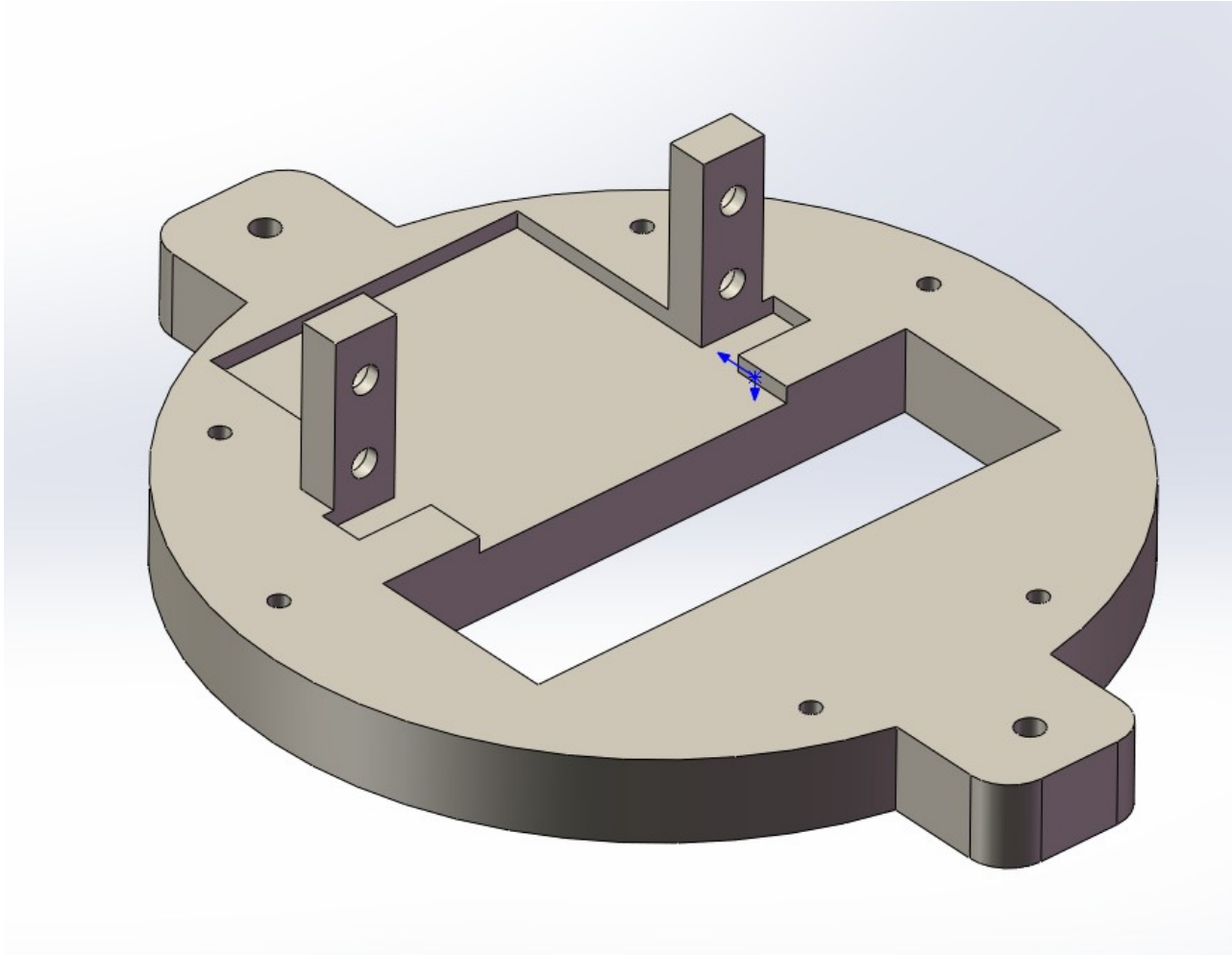
CAD Design - Assembly



Small Plane

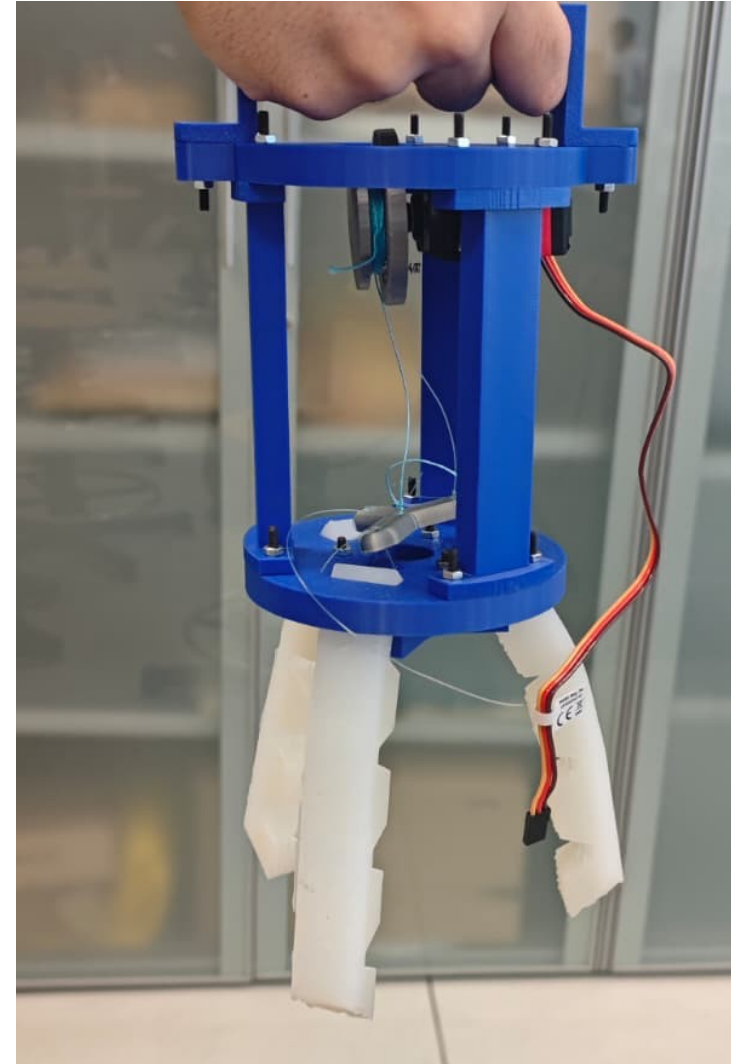
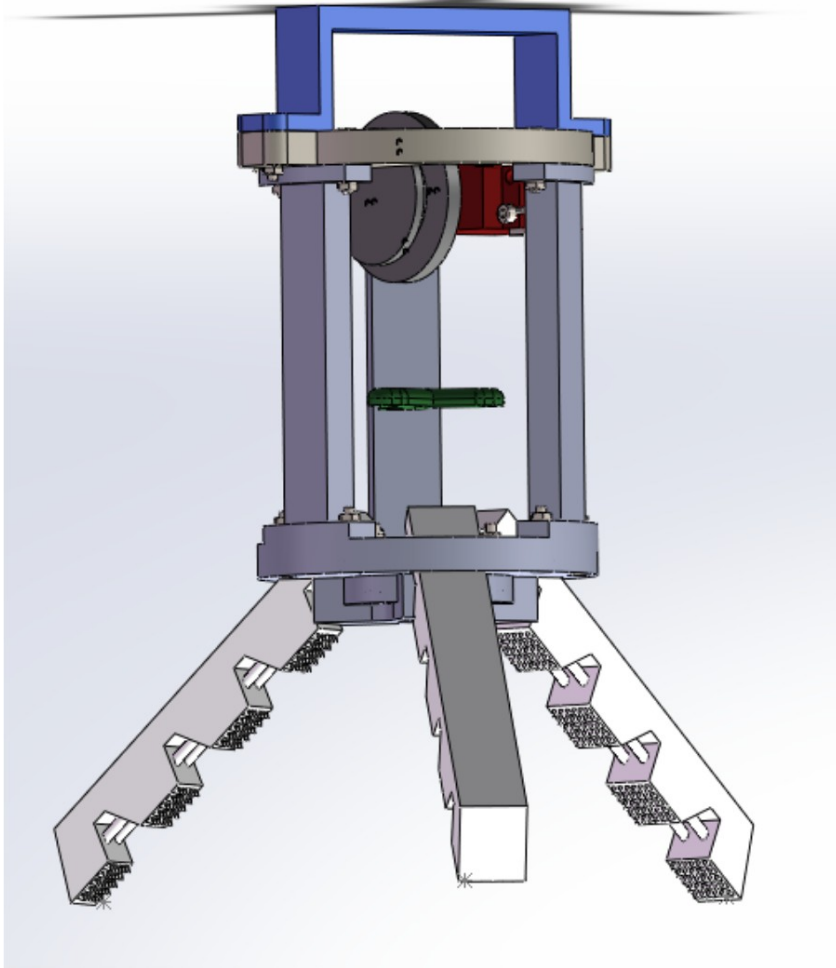


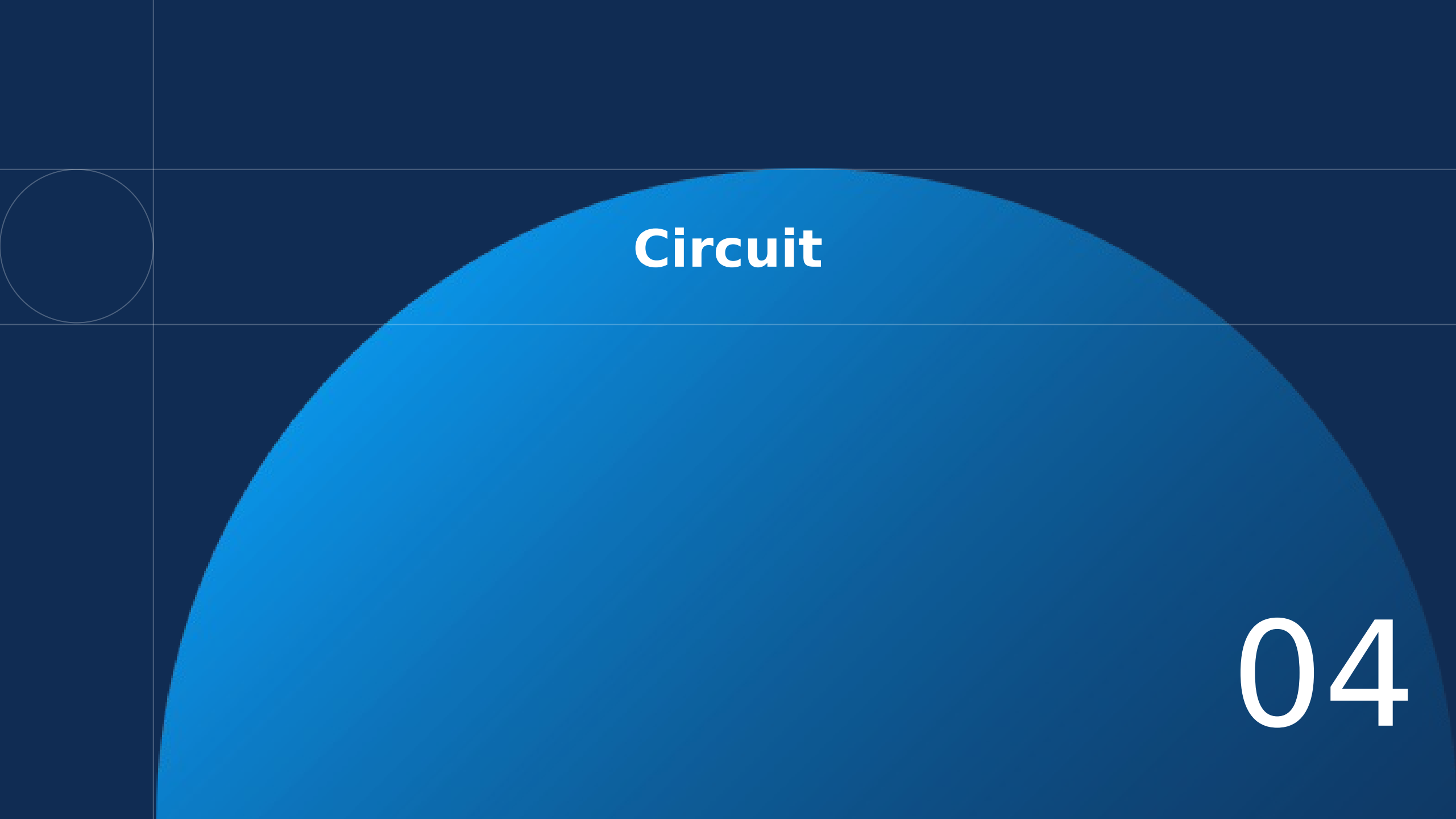
CAD Design - Base for Motor



Motor Base

CAD Design - Final Assembly

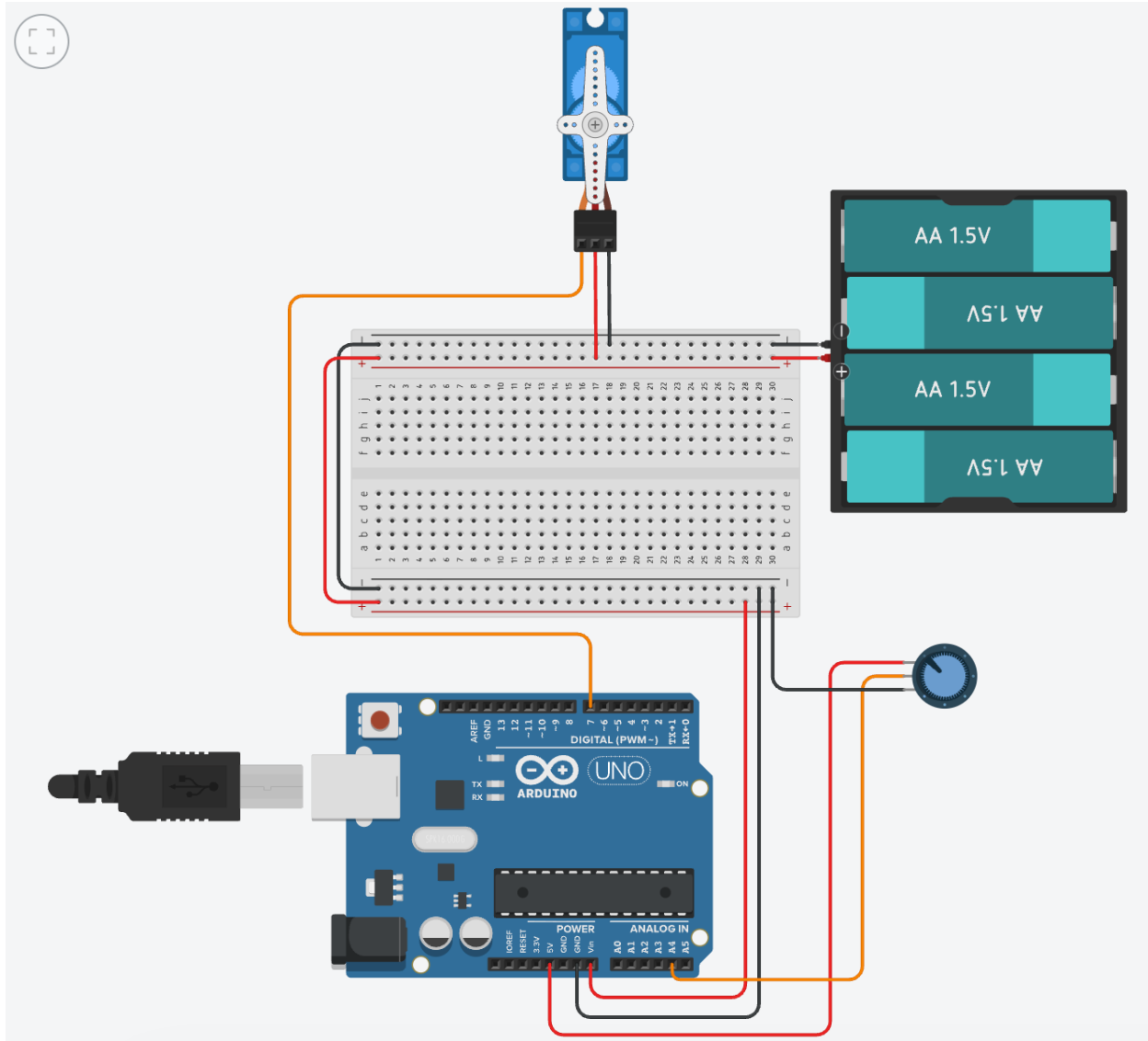




Circuit

04

CAD Design - Circuit



The Input (Potentiometer): This is the twistable knob. It acts as a variable resistor.

The Brain (Arduino): It constantly reads the voltage from the knob to figure out its position, translates that into a specific angle, and sends a control signal out to the motor.

The Actuator (Servo Motor): The motor receives the signal and physically rotates its arm to pull the tendon.

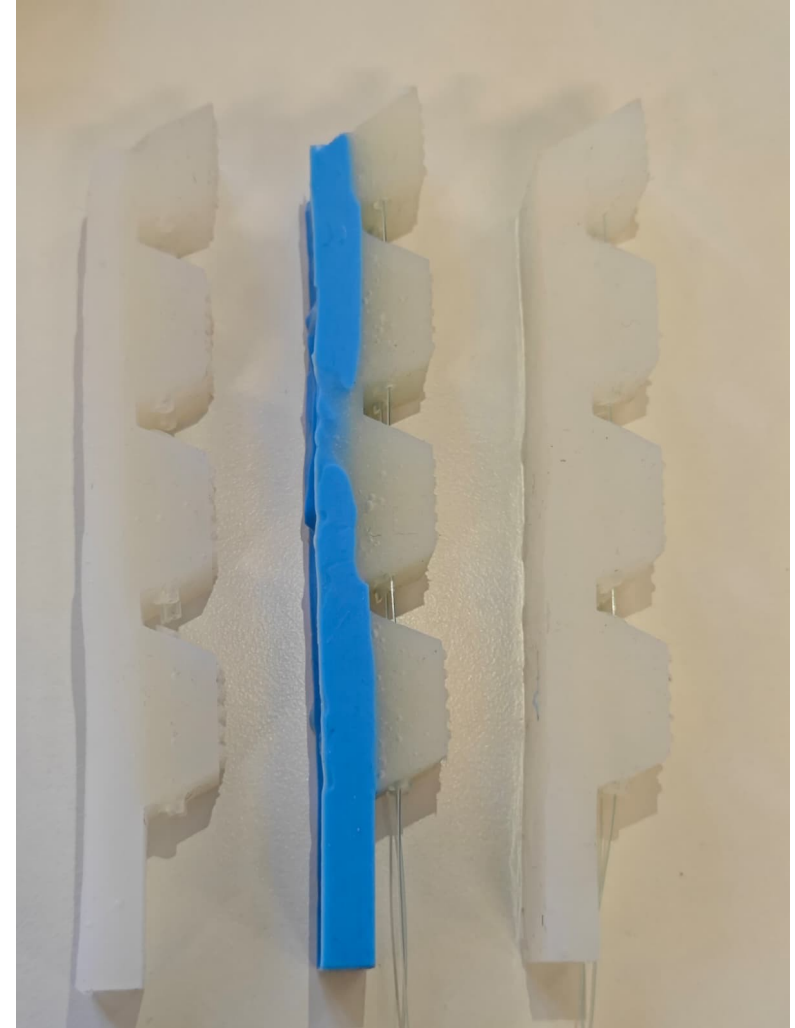
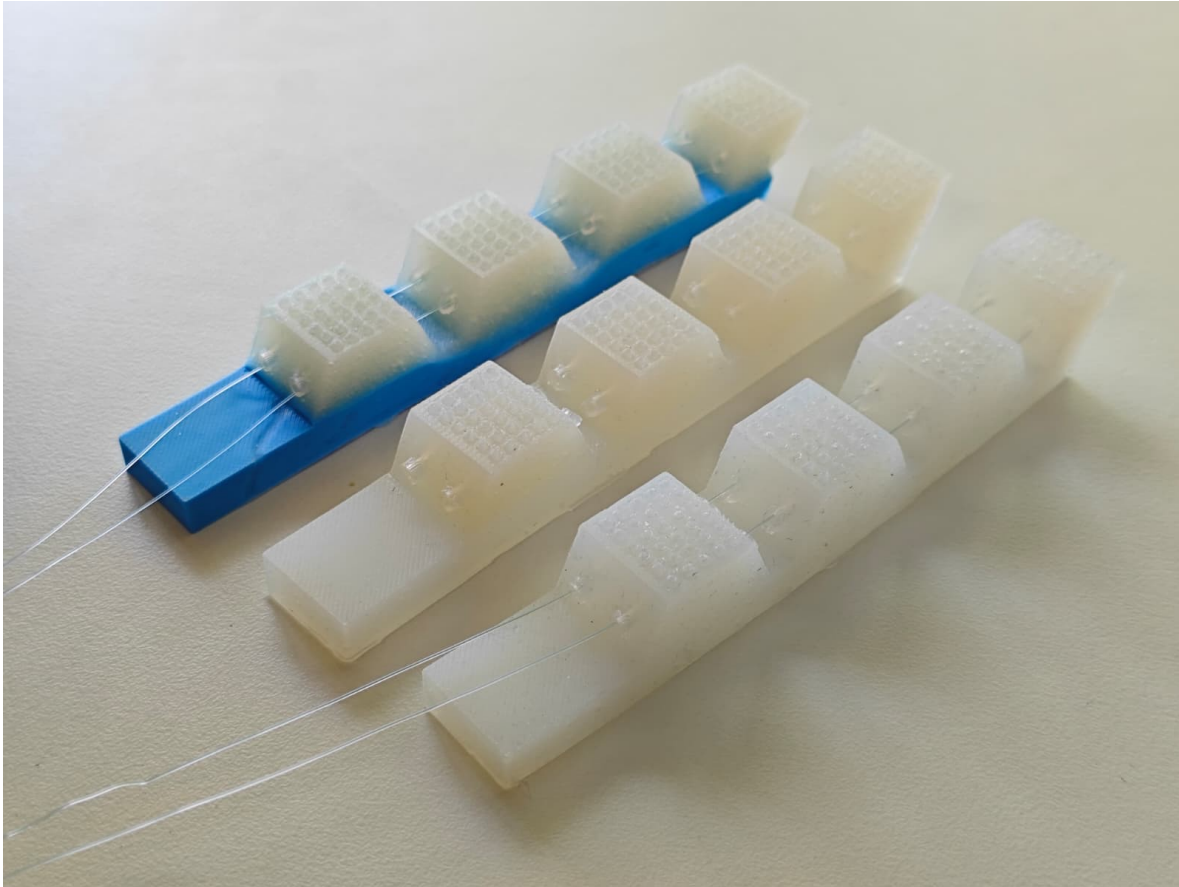
The Power Supply (Batteries): safely provides the power (6V) directly to the motor.



Materials and Methods

05

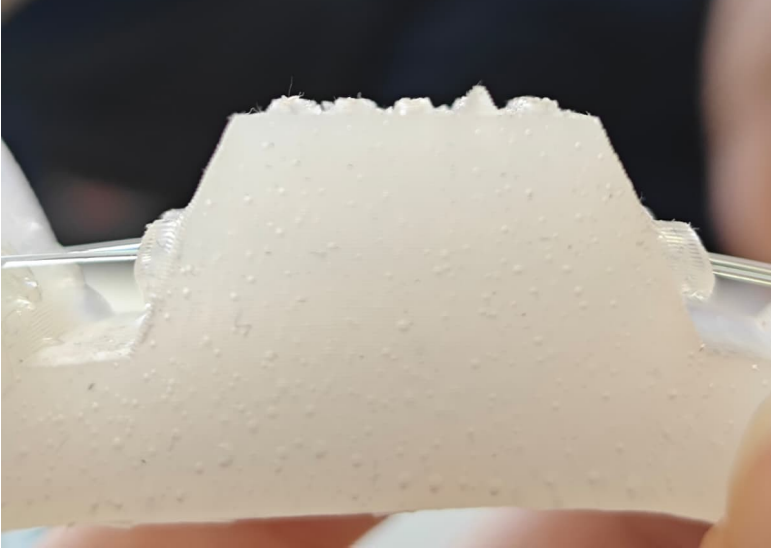
Material Choice



Comparing the Materials

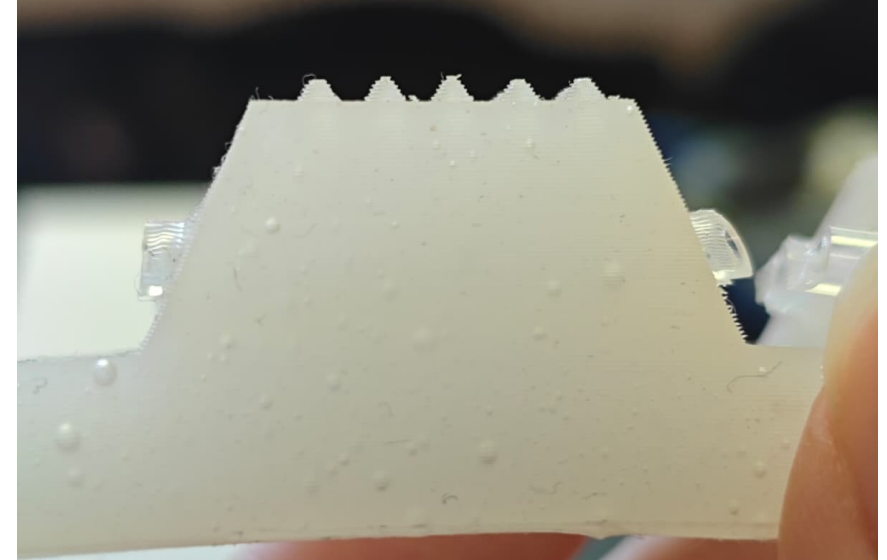
Property / Feature	Smooth-Sil Body	Dragon Skin Body
Material Stiffness	High stiffness (rigid/firm)	Medium stiffness (flexible)
Fin-Ray Flexing	Harder to bend; requires larger force to wrap around objects.	Bends smoothly to conform to the object's shape.
Tear Resistance	High	High
Behavior Under Load	Superior: Capable of lifting significantly heavier objects without buckling.	Holds its structure without tearing or collapsing.
Implementation Status	Promising for future heavy-lift iterations.	Chosen Baseline: Cast first, worked exceptionally well, and met core requirements.

Material Choice



Ecoflex 10

VS
▪



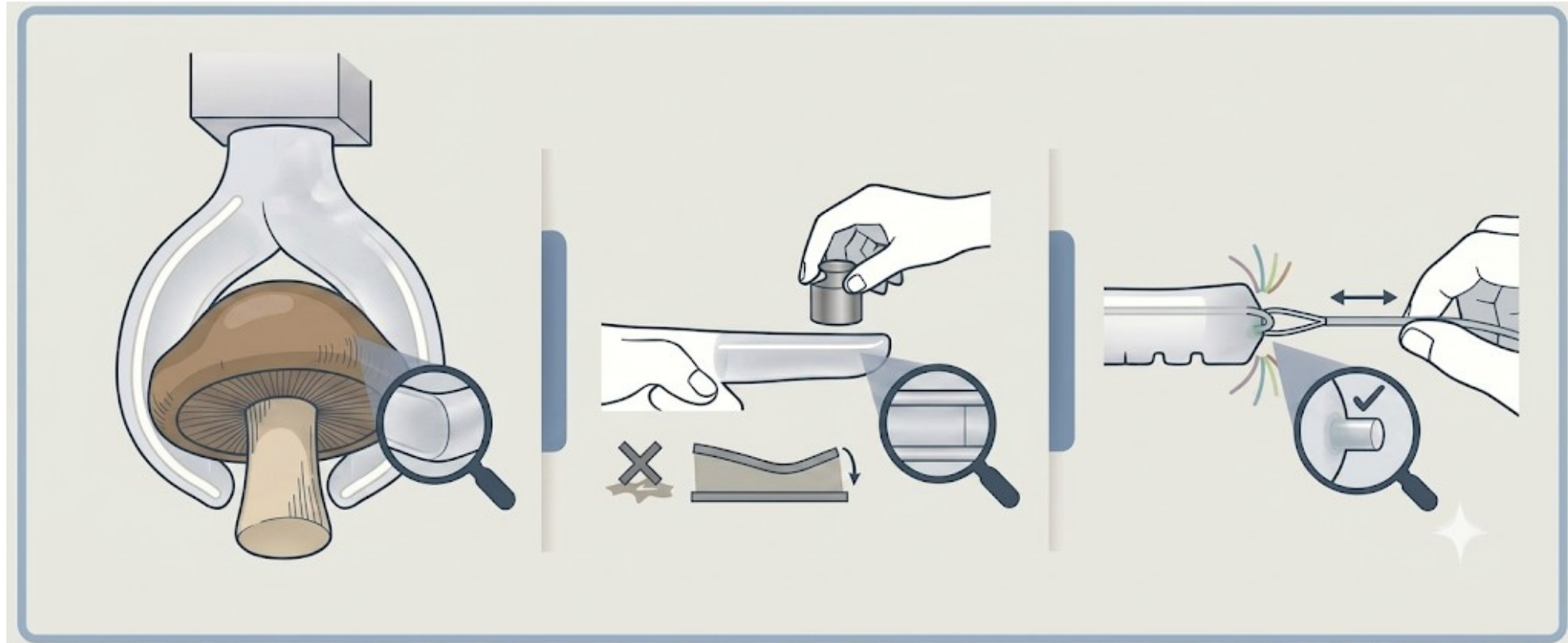
Ecoflex 00-35 Fast

As shown in the image, **Ecoflex 00-35 Fast captures the small gripping pattern much better** than Ecoflex 10, producing sharp, well-defined teeth.

Comparing the Materials

Property	Ecoflex 00-35 Fast	Ecoflex 10
Hardness (Shore)	00-35 (<i>Extremely soft, marshmallow-like</i>)	10 A (<i>Soft, but firm like a gummy bear or pencil eraser</i>)
Mixed Viscosity	Low (~11,000 cps)	Medium (~14,000 cps)
Cure Time	5 minutes	4 hours

Why we have a 2-layer design



No single material can solve all the biological constraints.

The Dragon Skin core allows the body to passively flex and wrap around objects. And its backbone prevents the finger from buckling under a payload. It effortlessly withstands the repeated stress. With Ultra-soft, high-friction Ecoflex 00-35 Fast skin, we achieve a highly durable finger that can securely, yet gently, grab objects.



Results

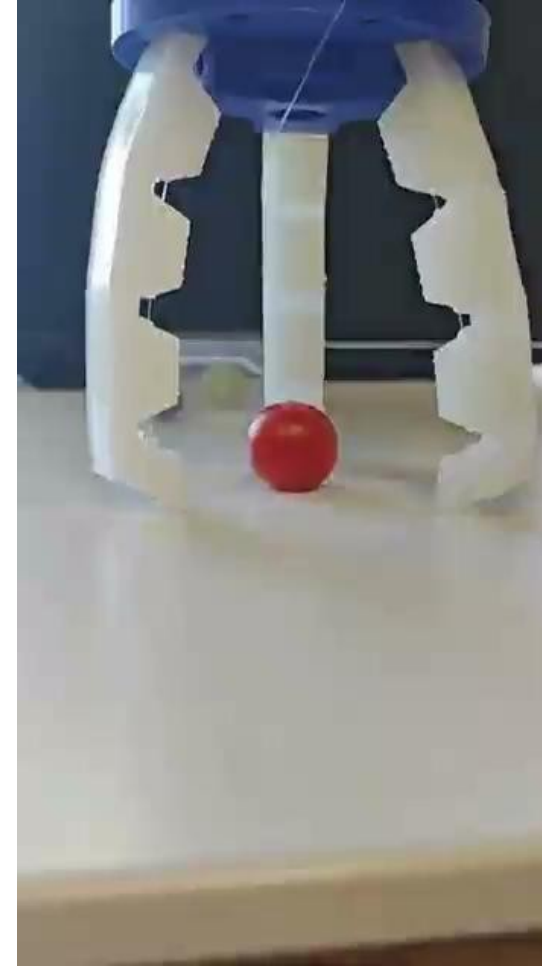
06

Experimental Demonstration Video



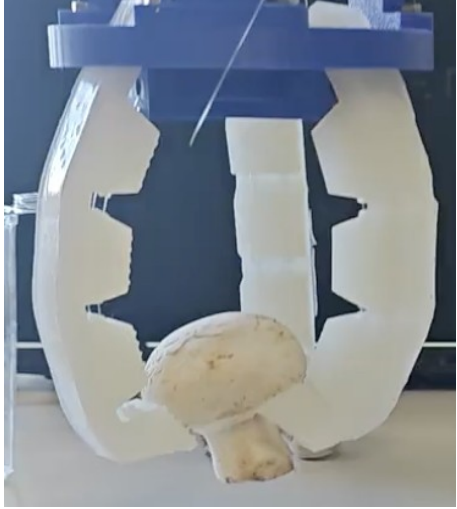
Mushroom

Success Attempts



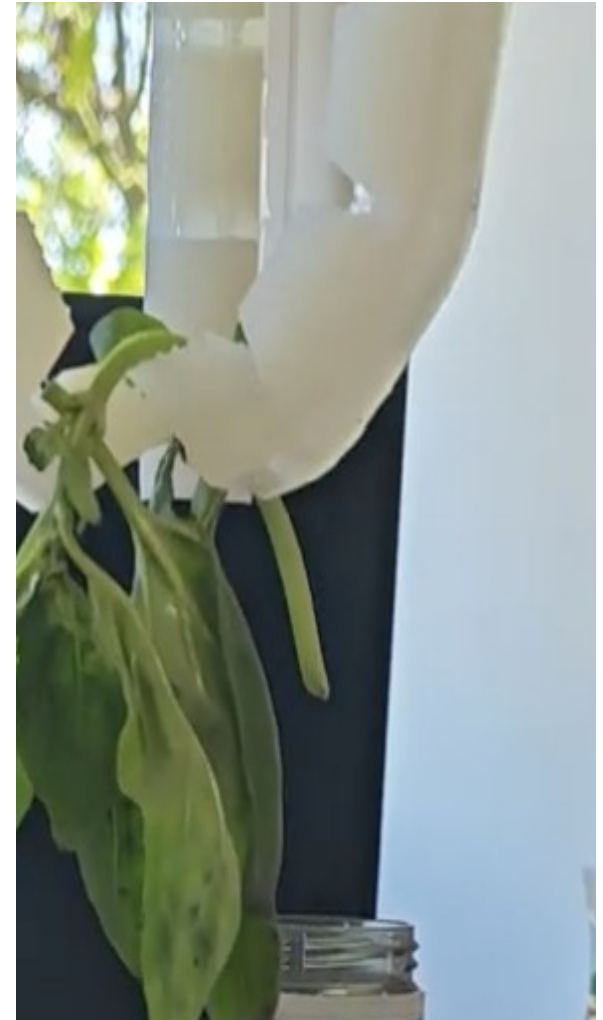
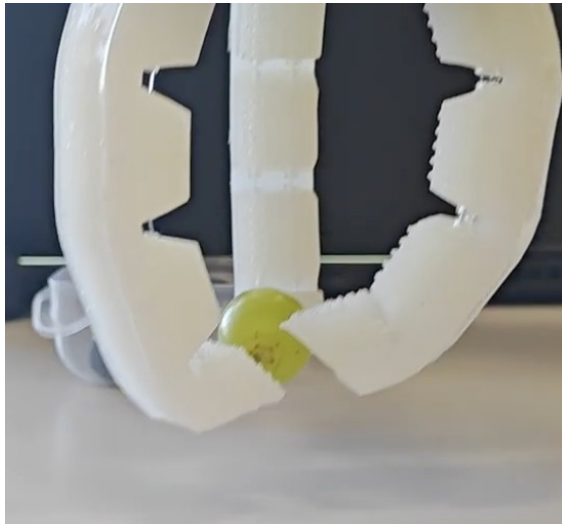
Failure Attempts

Performance of Grabbing - Target Object



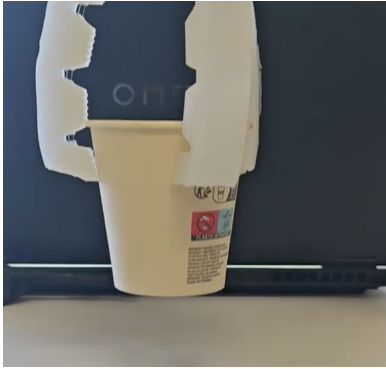
- **Fingers conform passively to mushroom**
- **No damage to the mushroom**
- **Stable holding**
- **The variable cross-section geometry works**

Performance of Grabbing - Other Object(Fruits & Daily Objects) 1 attempts



Performance of Grabbing - Other Object(Bottles & Cans)

1 attempts



● Water Bottle(142g)
Successful grasp and retention

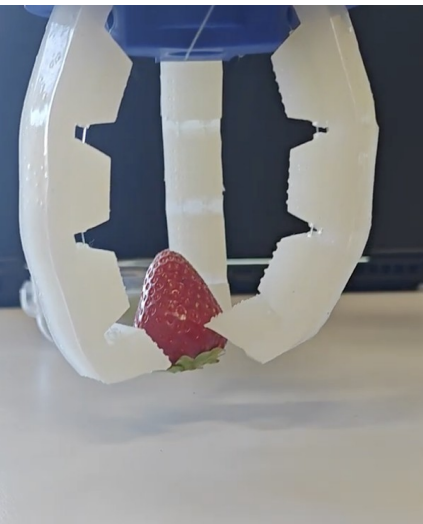
Performance of Grabbing - Other Object

2-4 attempts



- **Blueberry-2 times attempts**

- **Ketchup-4 times attempts**

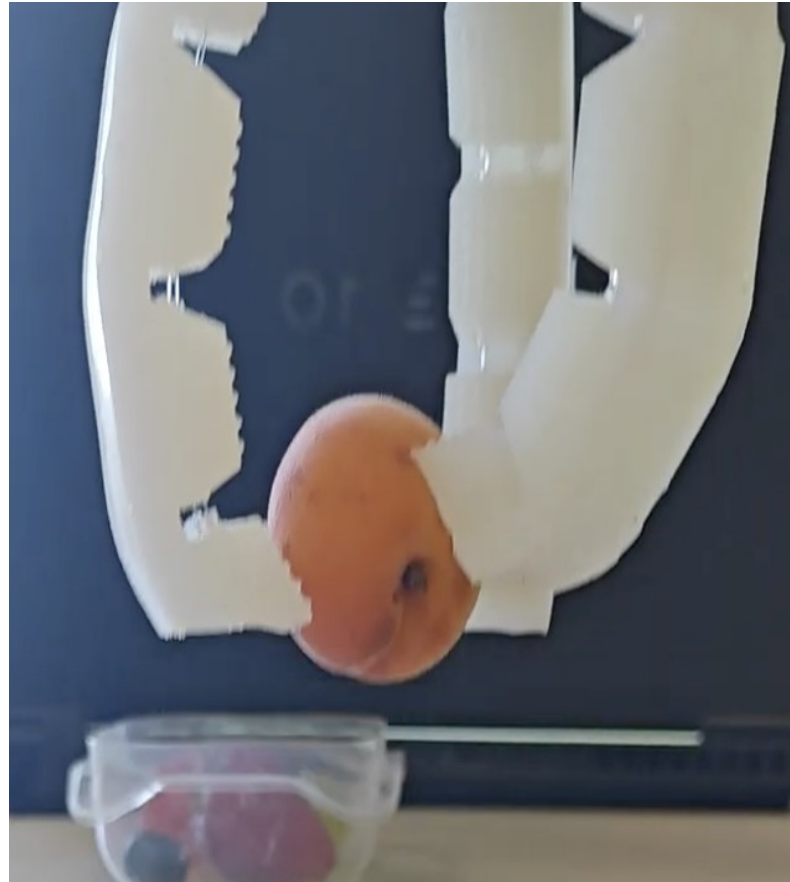


- **Strawberry — 2 attempts**



Performance of Grabbing - Other Object

5 attempts

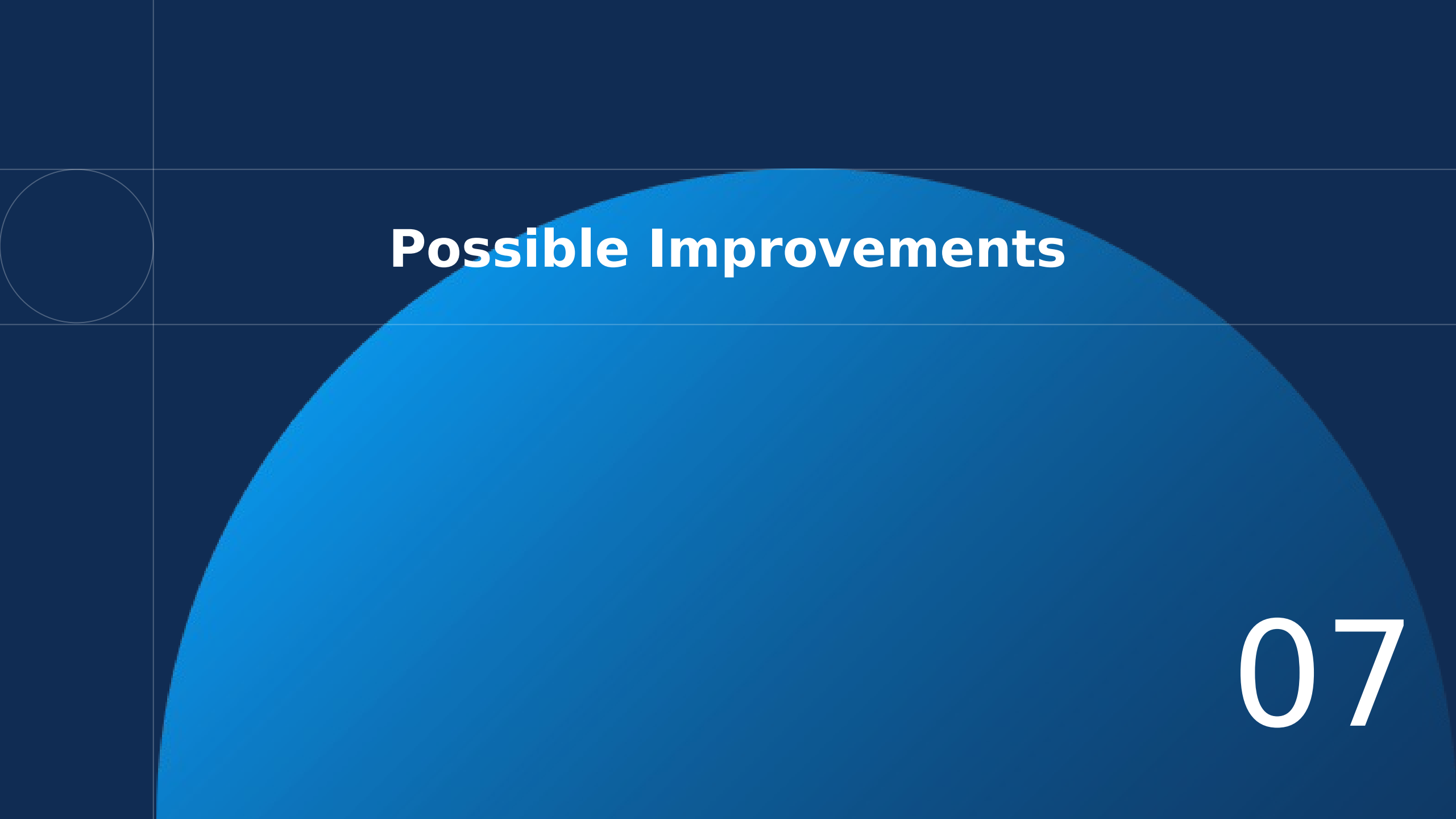


- **Cherry tomato-5 times attempts**
- **Apricot -5 times attempts**

Summary — Successful Cases

Items Name	Weight / g	Attempt Times
Mushroom	17	1
Cherry	10	1
Apricot	31	5
Strawberry	17	2
Basil	4	1
Cherry Tomato	11	5
Grape	7	1
Blueberry	3	2
Ketchup bottle	73	4

Items Name	Weight / g	Attempt Times
Water bottle	142	1
Tuna can	87	1
Adapter	86	1
Sunglass box	62	1
Coffee cup	4	1
Beverage can	14	1



Possible Improvements

07

Failure Reasons

- **Object too small for the aperture, so it falls through or sits in the dead zone between fingers**
- **The fingertip has no Dragon Skin behind it, so it cannot press hard and supply enough friction force to hold objects.**
- **Grip point above the center of mass, making unstable on rigid, tapered shapes**

Possible improvements

- **Close the finger dead zone. Use a smaller opening, or add soft webbing between the fingers, so small objects get caught instead of slipping into the gap.**
- **Strengthen the fingertip grip. Since the variable cross-section means we cannot add a stiff layer at the tip, use a deeper, higher-grip texture so it holds with less force, and tune the taper to keep a little more support at the tip.**
- **Improve the clamping force. Run the tendons through thin tubes so they lose less force, use a stiffer Smooth-Sil finger for rigid or heavy objects, and grip nearer the heaviest part.**



Thanks for listening!